Motherson Automotive & Technology engineering
Survery No 172/1-5,274/2,276/7,
Chengalpattu Sriperumbudur Road, Vadgal Village B Vallam Panchayat,,Vadagal,
Kanchipuram, Tamil Nadu, 602105

CUSTOMER NAME	MMLI PUNE
PROJECT NAME	S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1
PROJECT CODE	FMSPM00857
COMMISSION	ROBIS
PROJECT TYPE	WORK STATION
PLACE OF INSTALLATION	MMLI PUNE
RESPONSIBLE OF PROJECT	ROBIS
REVISION	01(Screw driver wiring updated)
DESIGNER	MANORAJ.S

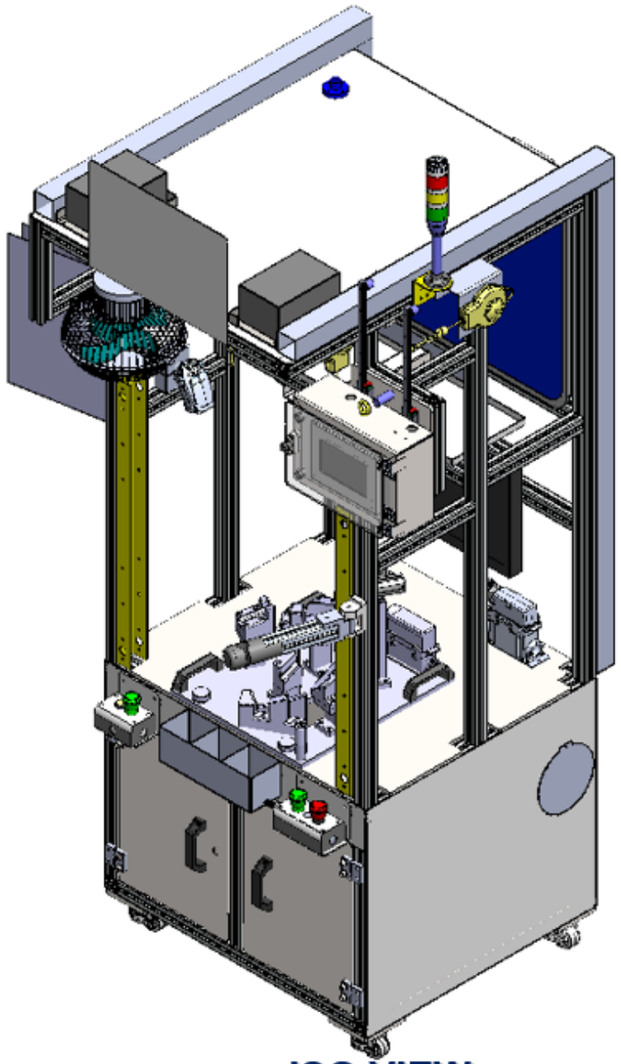
CREATED ON	27-07-2023	CHECKED BY			
LAST EDITED DATE	11-04-2024	CHECKED ON	24-07-2023	NUMBER OF PAGES	37

+CP/2

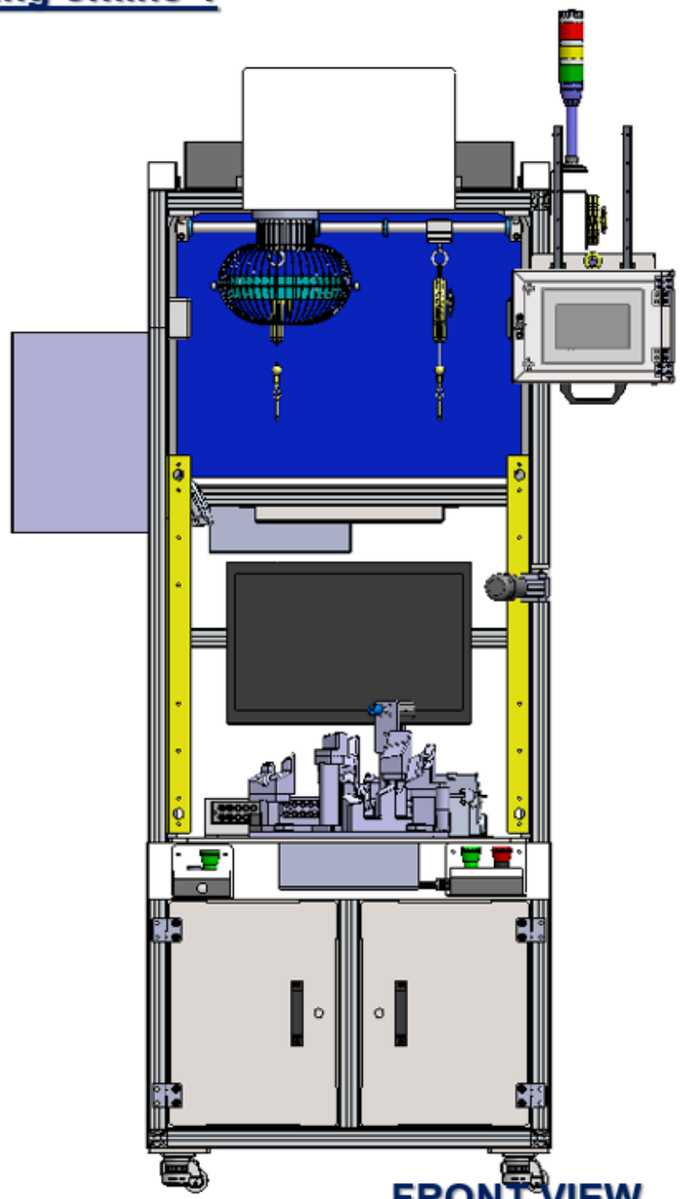
MACHINE OVER VIEW

FMSPM00857-S220 HL_ HV & LV Outer lens to bezel screwing offline-1

motherson



ISO VIEW



FRONT VIEW

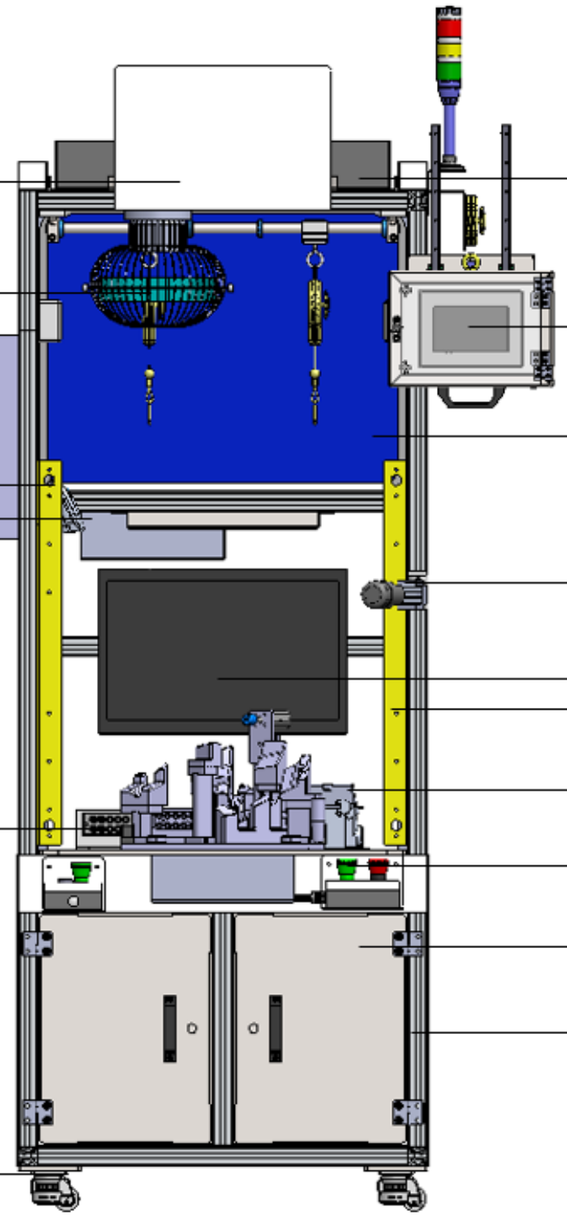
NOTE : PRINTER NOT CONSIDER

REVISION			Date	11-04-2024	ROBIS		Motherson Automotive & Technology engineering	MACHINE OVER VIEW		=
			Ed.	MANORAJ.SELVARAJ						+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1				FMSPM00857	Page 2
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MACHINE FRONT VIEW

motherson

FMSPM00857-COMPONENTS;



NAME PLATE

METALIC FAN (REMI MAKE)

A3 DOCUMENT FOLDER

SCANNER MOUNTING

PNEUMATIC PANEL

KDM10-06 PNEUMATIC CONNECTOR

GD-60 FOOT MASTER

SCREW DRIVER CONTROLLLER MOUNTING

HMI SLIDINGTYPE WITH ACRYLIC GUARD

PINUP BOARD-BLUE COLOUR

SCREW DRIVER PARKING

MONITOR

LIGHT CURTAIN(600mm RANGE)

24 PIN HARTING CONNECTOR

START/STOP PUSH BUTTONS(TECHNIK MAKE)

FRONT DOOR

BOTTOM STRUCTURE

CAMERA(OMRON - GIGE VISION AREA SCAN CAMERA , 12.0 MP STC MSC1242POE)

15

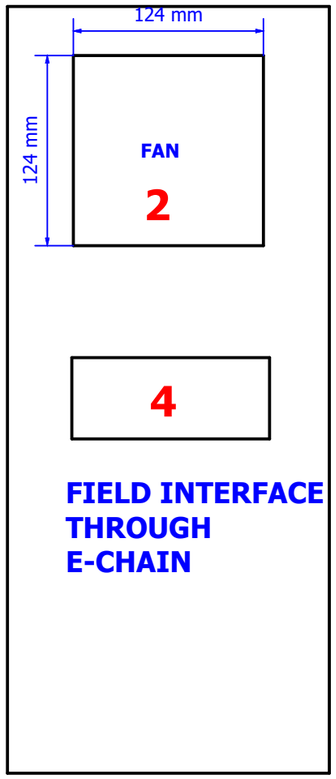
NOTE : PRINTER NOT CONSIDER

REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	MACHINE FRONT VIEW		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 3
Modification	Date	Name	Original		Replacement of	Replaced by			Page 3 / 37

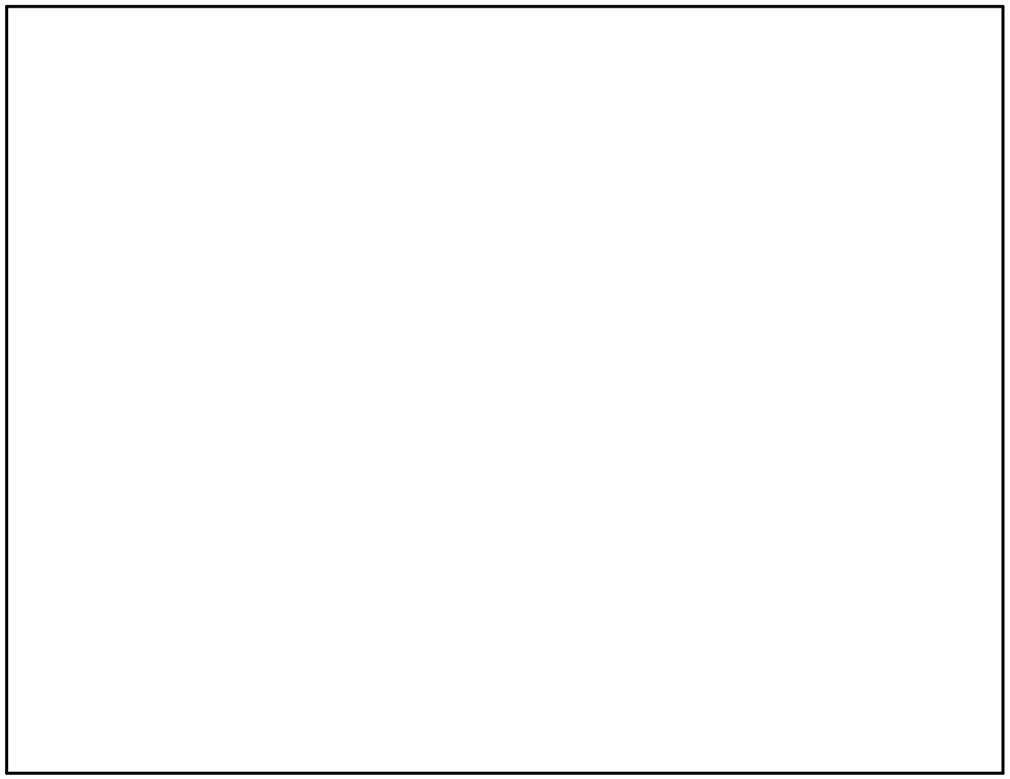
PANEL CUTOUT

CUSTOMIZED PANEL
H 500 X W 650 X D 210

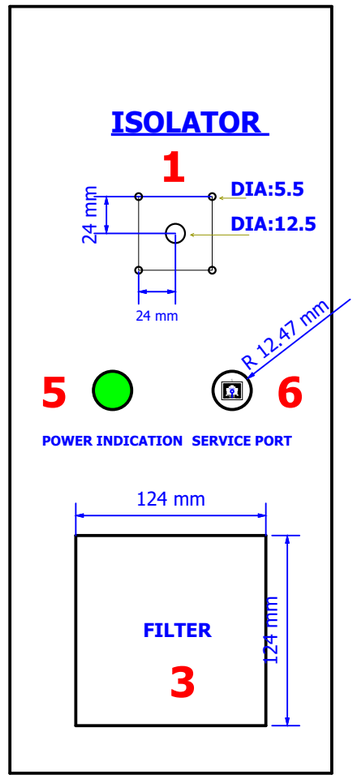
LEFT SIDE VIEW



FRONT SIDE VIEW



RIGHT SIDE VIEW



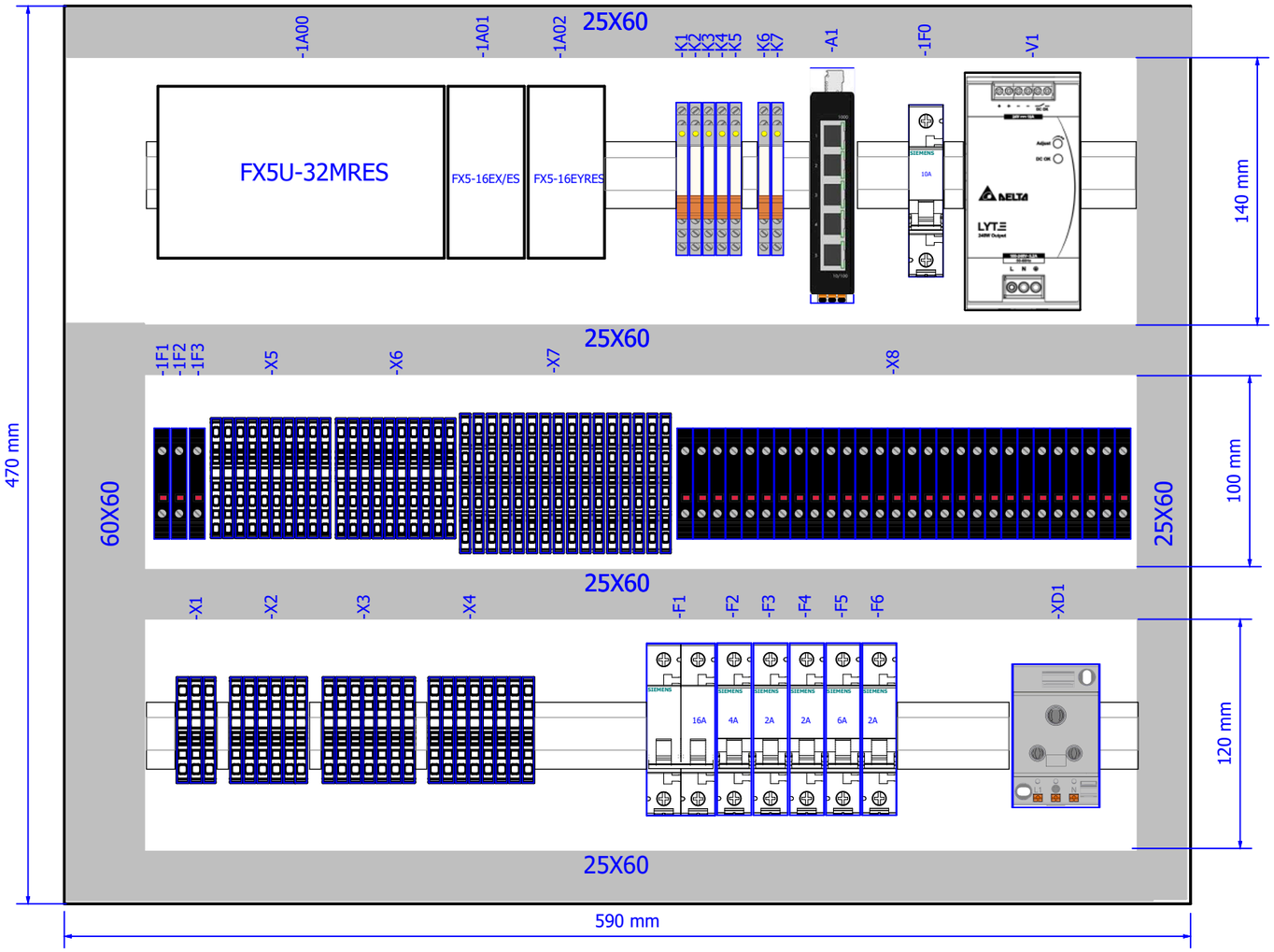
1	ISOLATOR CUTOUT
2	PANEL FAN CUTOUT
3	PANEL FILTER CUTOUT
4	FIELD INTERFACE CUTOUT
5	POWER INDICATION
6	SERVICE PORT

3 5

REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	PANEL CUTOUT		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 4
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LAYOUT DIAGRAM

-1A00	CAAAZ00000000077 PLC FX5U32MRES MITSUBHISHI	1
-1A01	CAAAZ00000000072 16POINT INPUT CARD FX5-16EX/ES -MITSUBHISHI	1
-1A02	CAAAZ00000000073 16POINTS OUTCARD :-FX5-16EYRES, MISTUBHISHI	1
-K1 TO -K7	BMELES0935 SLIM RELAY COIL 24VDC	7
-A1	BMELES1310 ETEHRNET SWITCH 100 MB/S	1
-1F0	BMELES1708 MCB SINGLE POLE 10A SIEMENS MAKE	1
-V1	BMELES1532 DELTA LITE POWER SUPPLY 10A ,1 Phase ,240W	1
-1F1	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	1
-1F2	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	1
-1F3	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	1
-X1	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	2
-X1	BMELES0595 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT YE/GN	1
-X2	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	6
-X3	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	7
-X4	BMELES0595 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT YE/GN	8
-X5	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	10
-X6	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	10
-X7	BMELES0366 TERMINAL BLOCK 2.5 SQ MM 2 IN 2 OUT	16
-X8	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	29
-F1	BMELES3072 SIEMENS MCB 2POLE 16AMPS 5SL62167RC	1
-F2	BMELES1706 MCB SINGLE POLE 4A SIEMENS MAKE	1
-F3	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
-F4	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
-F5	BMELES1707 MCB SINGLE POLE 6A SIEMENS MAKE	1
-F6	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
XD1	CAABA00000000433 804000 INDIAN 6A DIN RAIL POWER SOCKET	1



REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	LAYOUT DIAGRAM		=
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PROJECT ARCHITECTURE



CAAAZ00000000077
PLC FX5U32MRES
MITSUBHISHI



CAAAZ00000000072
16POINT INPUT CARD
MODEL :-FX5-16EX/ES
MITSUBHISHI



CAAAZ00000000073
16POINTS OUTCARD
MODEL:-FX5-16EYRES,
MAKE :- MISTUBHISHI



BMELES1540
Mushroom Red Turn
Teknic 1NC 2PSML4 01
(2AML4+S2)



TOWERLIGHT 2
STG RG24V BUZ MNT
BMELES0523



SENSORS
(CUSTOMER
SCOPE)



BMPNES5128
NEBU-M8G3-K-5-LE3

SCREW DRIVER 1



(CUSTOMER
SCOPE)



BMELES2309
Ethernet Switch
1000 MB/S



BMCONS0728
MOOTEK Industrial PC
MBOX-520-I5-M7-8265U



BMELES3308
VIEW SONIC
22"Touch Monitor



BMELES1552
MUSHROOM PB GREEN
Teknic 1NO 2PSM3 10
(2AM3+S1)



BMELES1552
MUSHROOM PB GREEN
Teknic 1NO 2PSM3 10
(2AM3+S1)



BMELES1542
NONIII. YELLOW PUSH
BUTTON, Teknic 1 NO
2PSF8 10 (2AF8+S1)



CAAAI00000000586
HMI MODEL : GS2107-WTBD
MITSUBHISH



BMELES3765
OMRON GIGE VISION
AREA SCAN CAMERA
12.0 MP STC MSC1242POE



BMELES3959
Data logic power
scanner PD9531



BMELES3765
OMRON GIGE VISION
AREA SCAN CAMERA
12.0 MP STC MSC1242POE

INPUT/OUTPUT

SAFETY

ETHERNET COMMUNICATION

RS 232 COMMUNICATION

RS 485 COMMUNICATION

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Modification	Date	Name	Original		Replacement of	Replaced by			Page 6 / 37

MAIN POWER DISTRIBUTION



-F1
BMELES3072
SIEMENS MCB 2POLE 16AMPS
5SL62167RC



-Q1
CAABA00000000274
16 A /3P /1WAY ISOLATOR
ROTARY SWITCH

-X1
BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

-W0
3Gx2.5 mm²
BMELES1940

BMELES3408
INDUSTRIAL
SOCKET LEGRAND

INDUSTRIAL SOCKET

POWER SUPPLY
1 PHASE 230V
50-60 Hz

-X2
BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

-X3
BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

-X4
BMELES0595
TERMINAL BLOCK
4.0 SQ MM 1 IN 1 OUT YE/GN

HMI PANEL EARTH

PNEUMATIC PANEL EARTH

PANEL EARTH

6 8

REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	MAIN POWER DISTRIBUTION		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1				FMSPM00857	Page 7
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POWER SUPPLY

-W27.1
BMELES1938
3G1.5 sqmm cable

L01 / 11.3 -P1
N01 / 11.3 -P1
PE01 / 11.3 -P1

BK/1 1.5 mm²
BK/2 1.5 mm²
GNYE 1.5 mm²

L01
N01
PE01

-F2
MCB SINGLE POLE
4 A SIEMENS MAKE
5SL6104RC;
BMELES1706

2.5 mm²
BK
L02

-V1
BMELES1532
DELTA LITE POWER SUPPLY
10A ,1 Phase ,240W

Weidmüller
PROeco
L(+) N(-) PE
Adjust
22-28 V
+ + - -
13 14

1.5 mm² DBU 24VDC
1.5 mm² DBU P0VDC
2.5 mm² GNYE PE-1

-1F0:2 12.2 / 24VDC
-X6:1 12.6 / P0VDC
-X4:1:B 7.7 / PE-1

1L1-2 / 7.3 -X2:2:B
1N1-2 / 7.5 -X3:2:B
PE-2 / 7.7 -X4:2:B

2.5 mm² BK 1L1-2

-F3
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE
5SL61027RC

BK/1 0.75 mm²
3L1-2

-W1
BMELES0075
3 G 0.75 mm²

0.75 mm² BK/1 2L1-2
0.75 mm² BK/2 1N1-2
0.75 mm² GNYE PE-2

-M1
BMELES0049
SK 3322107-
FAN AND FILTER UNIT

-LS
BMELES1669
AE PANEL DOOR
LIMIT SWITCH -
SZ 4127010

21
22

-W2
BMELES0075
3 G 0.75 mm²

BK/2 0.75 mm² 4L1-2

-H1
BMELES2527
1Feet 6W LED
TUBE LIGHT WITH
ALUMINIUM BATTERY (CW)

1N1-3 / 7.5 -X3:3:B
1 mm² BU 1N1-3

1L1-3 / 7.3 -X2:3:B
2.5 mm² BK 1L1-3

-F4
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE
5SL61027RC

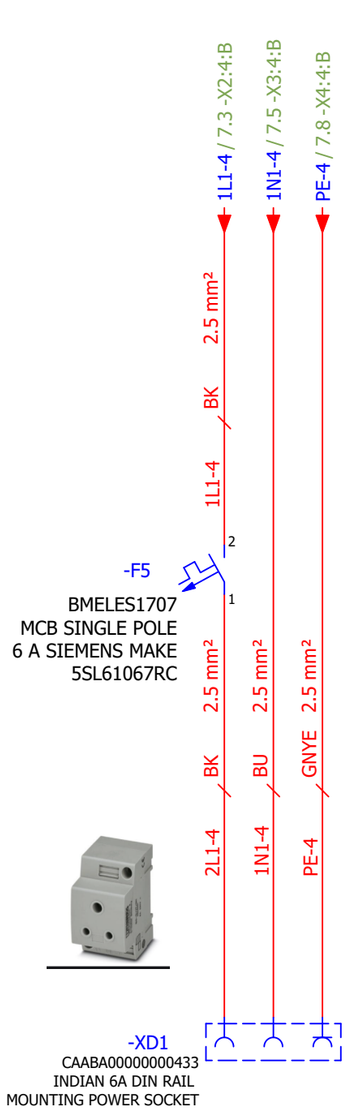
1.5 mm² BK 2L1-3

-K6:11 26.1 / 2L1-3

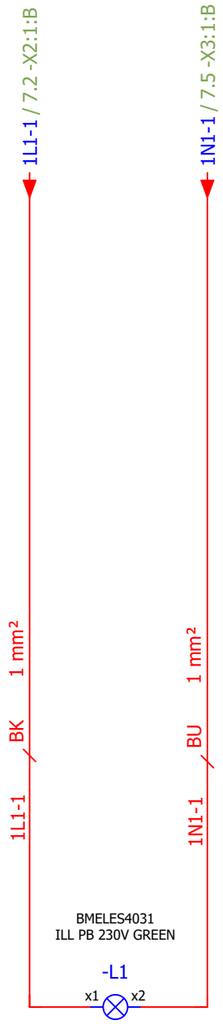
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REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	POWER SUPPLY		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
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Modification	Date	Name	Original		Replacement of	Replaced by				Page 8 / 37

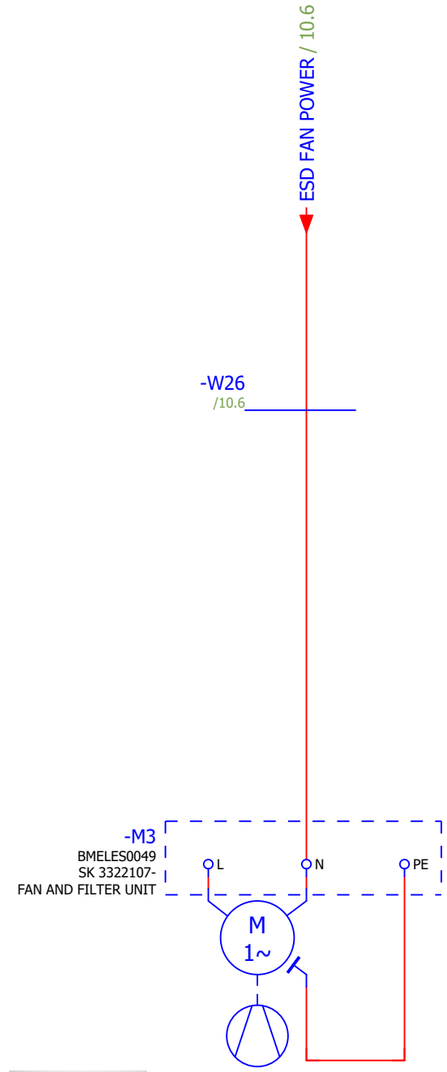
INDICATION LAMP



POWER SOCKET



230V GREEN
POWER INDICATION
LAMP



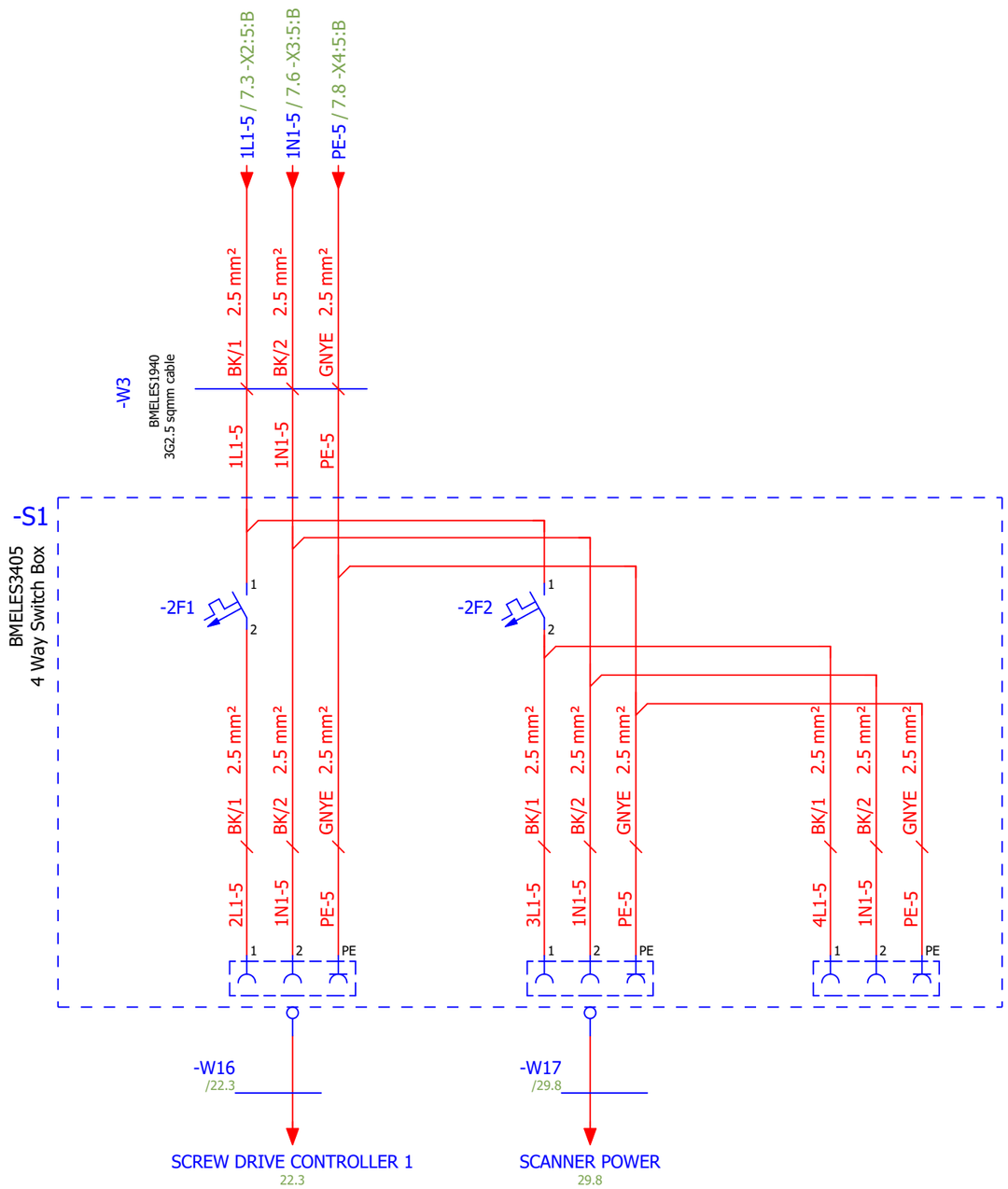


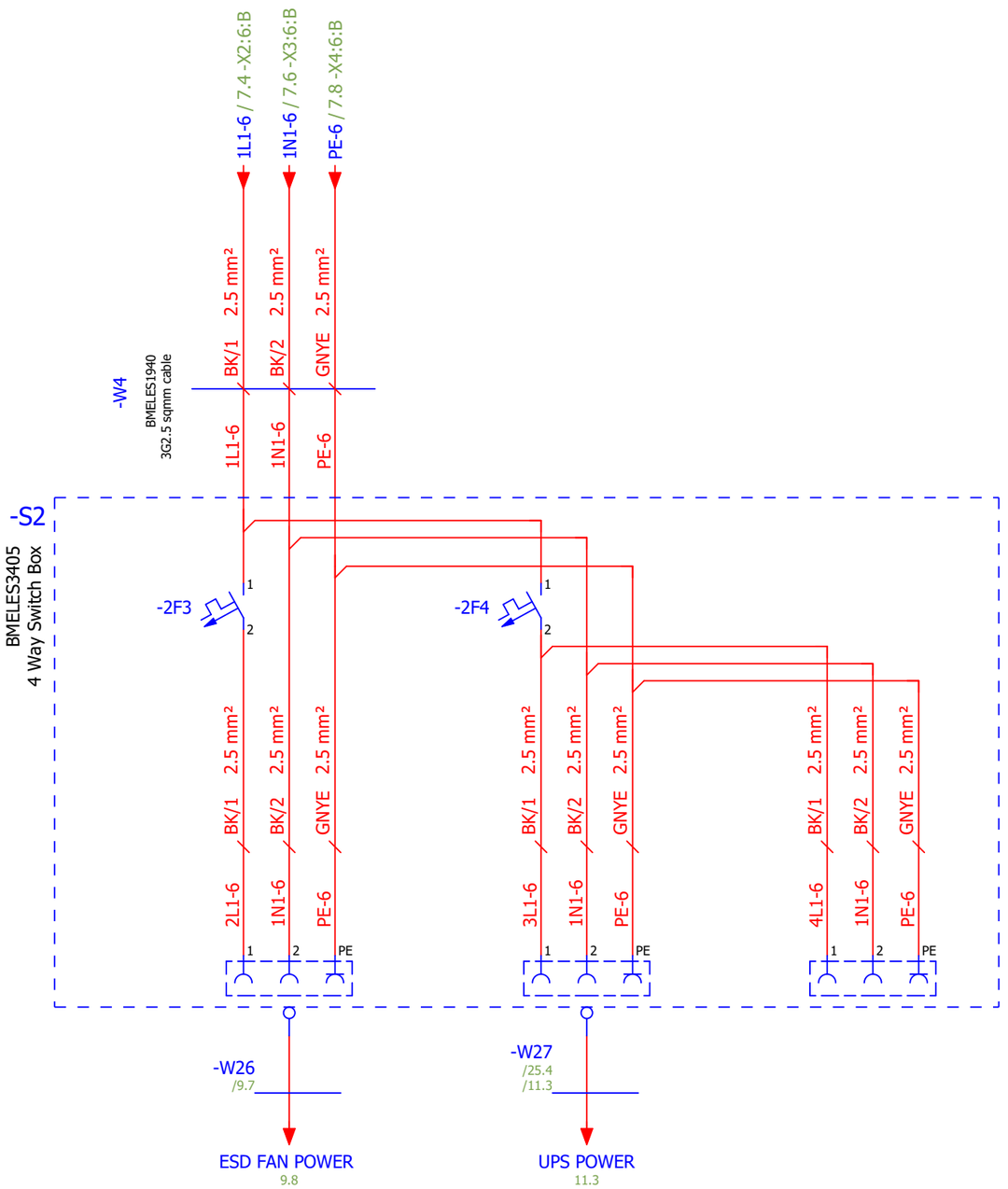
ESD FAN (CUSTOMER SCOPE)

8 10

REVISION			Date	11-04-2024	ROBIS		Motherson Automotive & Technology engineering	INDICATION LAMP	=
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SOCKET INTERFACE





REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	SOCKET INTERFACE		=
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Modification	Date	Name	Original		Replacement of	Replaced by			Page 10 / 37

UPS INTERFACE



UPS POWER / 10.7 /

-W27

-UPS
BMELES2946
APC 1.1kva Line Interactive
UPS with Built In Battery Backup

L 3 4

L 1 2 | L 1 2 | L 1 2 | L 1 2 | L 1 2

-P1
BMELES3265

-W27.1
BMELES1938
3G1.5 sqmm cable

BK/1 1.5 mm² L01
BK/2 1.5 mm² N01
GNYE 1.5 mm² PE01

-W27.2
POWER CABLE

MONITOR POWER
25.4

-F2:2 8.1 / L01
-V1:N(-) 8.1 / N01
-V1:PE 8.1 / PE01

TO SMPS

TO MONITOR

10

12

REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	UPS INTERFACE		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1				FMSPM00857	Page 11
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DISTRIBUTION 24VDC TERMINAL BLOCK

P24VDC1 / 12.2 -X5:1
P24VDC14 / 12.4 -X5:7
P24VDC16 / 12.4 -X5:8
24VDC / 8.1 -V1:+
P0VDC / 8.1 -V1:-



-1F0
BMELES1708
MCB SINGLE POLE 10A
SIEMENS MAKE
5SL61107RC

BMELES0584
TB4+HESILED
24 (5X20) EI

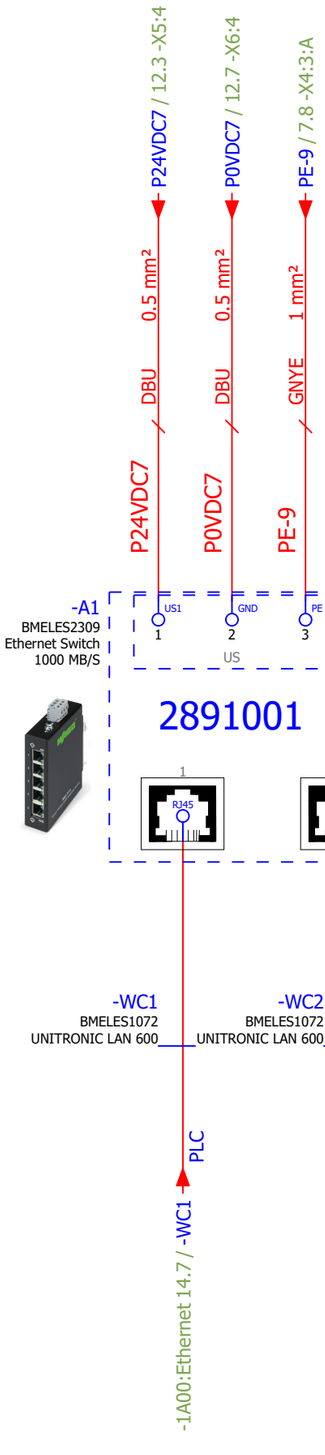
-1F1 2A
-1F2 2A
-1F2 2A

BMELES0364
TERMINAL BLOCK
2.5 SQ MM 1 IN 2 OUT

-X5
-X6

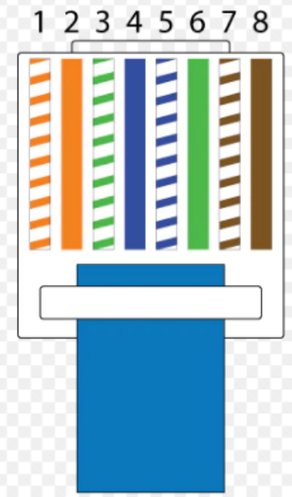
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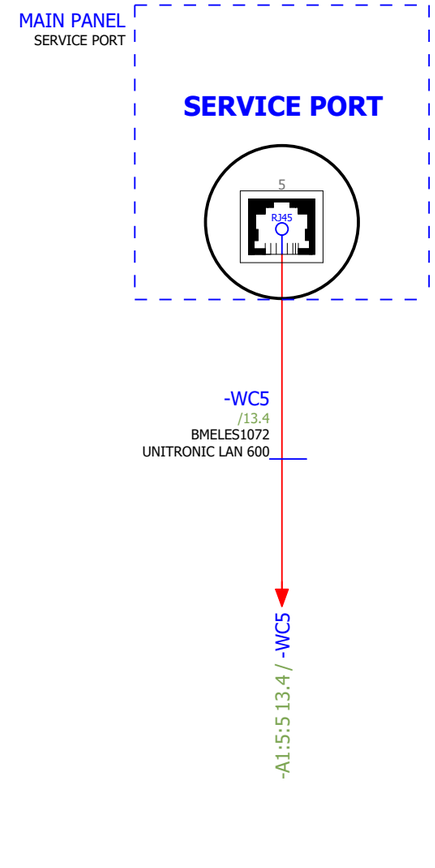
COMMUNICATION



RJ45 Pinout T-568B

1. White Orange
2. Orange
3. White Green
4. Blue
5. White Blue
6. Green
7. White Brown
8. Brown





REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	COMMUNICATION		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1				FMSPM00857	Page 13
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MAIN PLC POWER SUPPLY



-F6
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE
5SL61027RC

1L1-7 / 7.3 -X2:2:A
1N1-7 / 7.6 -X3:7:B
PE-7 / 7.9 -X4:7:B

1L1-7 — BK — 2.5 mm² — 2 — 1 — 2.5 mm² — BK — 2L1-7
1N1-7 — 2.5 mm² — BU — 1N1-7
PE-7 — 2.5 mm² — GNYE — PE-7

WC1 / 13.1 -A1:1:1

-WC1
/13.1
/28.1
BMELES1072
2170144-UNITRONIC
LAN 600S/FTP CAT7Y



-1A00
CAAAZ00000000077
PLC
FX5U32MRES
MAKE :- MITSUBHISHI

L	N	PE	SG (GND)	SDB (TXD-)	SDA (TXD+)	RDB (RXD-)	RDA (RXD+)	V1+ (PEW0)	V2+ (PEW2)	V-' (PEW4)	V+ (PAW0)	V- (PAW2)	Ethernet

Ethernet port

MITSUBISHI ELECTRIC
Mitsubishi FX5U-32MR-ES 16DI/16DO Part 1 of 5

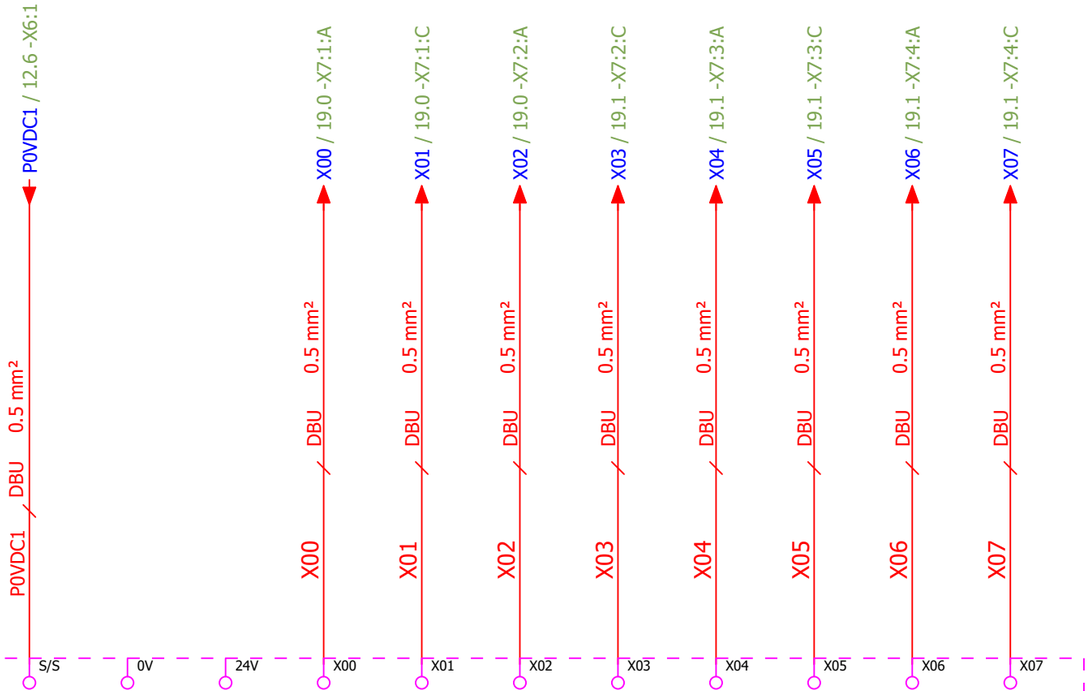
13 15

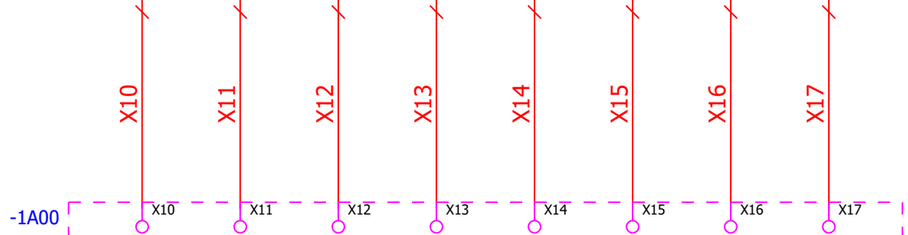
REVISION			Date	11-04-2024	ROBIS		Motherson Automotive & Technology engineering	MAIN PLC POWER SUPPLY		=	
			Ed.	MANORAJ.SELVARAJ						+ CP	
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Modification	Date	Name	Original		Replacement of	Replaced by				Page	14 / 37

MAIN INPUT MODULE



-1A00
CAAAZ00000000077
PLC FX5U32MRES
MAKE :- MITSUBHISHI





REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	MAIN INPUT MODULE	=	
			Ed.	MANORAJ.SELVARAJ				+ CP	
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 15
Modification	Date	Name	Original		Replacement of	Replaced by			Page 15 / 37

MAIN OUTPUT MODULE



-1A00
CAAAZ00000000077
PLC FX5U32MRES
MAKE :- MITSUBHISHI

Mitsubishi FX5U-32MR-ES 16DI/16DO Part 4 of 5

MITSUBISHI ELECTRIC

	TOWER LAMP YELLOW	TOWER LAMP BUZZER			MACHINE LIGHT			SCREW DRIVER RESET	
TOWER LAMP GREEN		TOWER LAMP RED			SHUTOFF VALVE			SCREW DRIVER ENABLE	
COM0	A0.0 Y00	A0.1 Y01	A0.2 Y02	A0.3 Y03	COM1	A0.4 Y04	A0.5 Y05	A0.6 Y06	A0.7 Y07

0.5 mm² DBU P24VDC2 → -X5:1 12.2 / P24VDC2

0.5 mm² DBU Y00 → -X8:1:A 20.1 / Y00

0.5 mm² DBU Y01 → -X8:2:A 20.1 / Y01

0.5 mm² DBU Y02 → -X8:3:A 20.1 / Y02

0.5 mm² DBU Y03 → -X8:4:A 20.1 / Y03

0.5 mm² DBU P24VDC2 → -X8:5:A 20.2 / Y04

0.5 mm² DBU Y04 → -X6:A1 26.1 / Y05

0.5 mm² DBU Y05 → -K1:A1 21.0 / Y06

0.5 mm² DBU Y06 → -K2:A1 21.2 / Y07

0.5 mm² DBU Y07

15

-1A00

Mitsubishi FX5U-32MR-ES 16DI/16DO Part 5 of 5

MITSUBISHI ELECTRIC

SPARE		CLAMPING CYL 1 REV					SPARE VALVE		SPARE VALVE
	SPARE		CLAMPING CYL 1 FWD					SPARE VALVE	SPARE VALVE
COM2	A1.0 Y10	A1.1 Y11	A1.2 Y12	A1.3 Y13	COM3	A1.4 Y14	A1.5 Y15	A1.6 Y16	A1.7 Y17

0.5 mm² DBU P24VDC3 → -X5:2 12.2 / P24VDC3

0.5 mm² DBU Y10 → -X8:6:A 20.2 / Y10

0.5 mm² DBU Y11 → -X8:7:A 20.2 / Y11

0.5 mm² DBU Y12 → -X8:8:A 20.2 / Y12

0.5 mm² DBU Y13 → -X8:9:A 20.2 / Y13

0.5 mm² DBU P24VDC3 → -X8:10:A 20.3 / Y14

0.5 mm² DBU Y14 → -X8:11:A 20.3 / Y15

0.5 mm² DBU Y15 → -X8:12:A 20.3 / Y16

0.5 mm² DBU Y16 → -X8:13:A 20.3 / Y17

0.5 mm² DBU Y17

17

REVISION			Date	11-04-2024	ROBIS		Motherson Automotive & Technology engineering	MAIN OUTPUT MODULE	=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 16
Modification	Date	Name	Original		Replacement of	Replaced by			Page 16 / 37

EXPANSION INPUT MODULE



P0VDC2 / 12.6 -X6:1 — DBU 0.5 mm² — P0VDC2

X20 / 19.3 -X7:9:A — DBU 0.5 mm² — X20

X21 / 19.3 -X7:9:C — DBU 0.5 mm² — X21

X22 / 19.3 -X7:10:A — DBU 0.5 mm² — X22

X23 / 19.3 -X7:10:C — DBU 0.5 mm² — X23

X24 / 19.3 -X7:11:A — DBU 0.5 mm² — X24

X25 / 19.3 -X7:11:C — DBU 0.5 mm² — X25

X26 / 19.3 -X7:12:A — DBU 0.5 mm² — X26

X27 / 19.4 -X7:12:C — DBU 0.5 mm² — X27

-1A01
/17.5

CAAAZ00000000072
16POINTINPUTCARD
MODEL :-FX5-16EX/ES
MAKE :-MITSUBHISHI

S/S	X20	X21	X22	X23	X24	X25	X26	X27
	E2.0	E2.1	E2.2	E2.3	E2.4	E2.5	E2.6	E2.7
	FIXTURE SPARE	FIXTURE SPARE	FIXTURE SPARE	FIXTURE SPARE	FIXTURE SPARE	FIXTURE SPARE	HIGH VARIANT	MID VARIANT

MITSUBISHI ELECTRIC

Mitsubishi FX5U-EX/ES 16DI Part 1 of 2

X30 / 19.4 -X7:13:A — DBU 0.5 mm² — X30

X31 / 19.4 -X7:13:C — DBU 0.5 mm² — X31

X32 / 19.4 -X7:14:A — DBU 0.5 mm² — X32

X33 / 19.4 -X7:14:C — DBU 0.5 mm² — X33

X34 / 19.4 -X7:15:A — DBU 0.5 mm² — X34

X35 / 19.5 -X7:15:C — DBU 0.5 mm² — X35

X36 / 19.5 -X7:16:A — DBU 0.5 mm² — X36

X37 / 19.5 -X7:16:C — DBU 0.5 mm² — X37

-1A01
/17.1

X30	X31	X32	X33	X34	X35	X36	X37
E3.0	E3.1	E3.2	E3.3	E3.4	E3.5	E3.6	E3.7
LOW VARIANT	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE

MITSUBISHI ELECTRIC

Mitsubishi FX5U-EX/ES 16DI Part 2 of 2

REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	EXPANSION INPUT MODULE		=
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EXPANSION OUTPUT MODULE



-1A02
/18.5
CAAAZ00000000073
16POINTS OUTCARD
-FX5-16EYRES, MAKE
MISTUBHISHI

Mitsubishi FX5-16EYR-ES 16DO Part 1 of 2

MITSUBISHI ELECTRIC

SPARE VALVE SPARE SPARE SPARE

SPARE VALVE CAMERA LIGHT ON SPARE SPARE

+V4 Y20 Y21 Y22 Y23 Y24 Y25 Y26 Y27

0.5 mm² DBU

P24VDC4

Y20 Y21 Y22 Y23 Y24 Y25 Y26 Y27

-X8:14:A 20.4 / Y20
-X8:15:A 20.4 / Y21
-X8:16:A 20.4 / Y22
-X8:17:A 20.4 / Y23
-X8:18:A 20.4 / Y24
-X8:19:A 20.5 / Y25
-X8:20:A 20.5 / Y26
-X8:21:A 20.5 / Y27

-X5:2 12.3 / P24VDC4

-1A02
/18.1

Mitsubishi FX5-16EYR-ES 16DO Part 2 of 2

MITSUBISHI ELECTRIC

SPARE SPARE SPARE SPARE

SPARE SPARE SPARE SPARE SPARE

+V6 Y30 Y31 Y32 Y33 Y34 Y35 Y36 Y37

0.5 mm² DBU

P24VDC5

Y30 Y31 Y32 Y33 Y34 Y35 Y36 Y37

-X8:22:A 20.5 / Y30
-X8:23:A 20.5 / Y31
-X8:24:A 20.6 / Y32
-X8:25:A 20.6 / Y33
-X8:26:A 20.6 / Y34
-X8:27:A 20.6 / Y35
-X8:28:A 20.7 / Y36
-X8:29:A 20.7 / Y37

-X5:3 12.3 / P24VDC5

17 19

REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	EXPANSION OUTPUT MODULE		=
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INPUT TERMINAL DISTRIBUTION

BMELES2970
TERMINAL BLOCK CXDL2.5
2.5 SQ MM 2 IN 2 OUT
-X7-

-PB1-S1:12 30.5 / X00	X00	BK/4	0.75 mm²	1	X00	DBU	0.5 mm²	X00 / 15.2 -1A00:X00
-PB2-S4:14 30.1 / X01	X01	BK/3	0.75 mm²		X01	DBU	0.5 mm²	X01 / 15.3 -1A00:X01
-PB1-S2:14 30.4 / X02	X02	BK/3	0.75 mm²	2	X02	DBU	0.5 mm²	X02 / 15.3 -1A00:X02
-PB2-S3:14 30.2 / X03	X03	BK/4	0.75 mm²		X03	DBU	0.5 mm²	X03 / 15.3 -1A00:X03
-PSC 28.6 / X04	X04	BK/2	0.75 mm²	3	X04	DBU	0.5 mm²	X04 / 15.4 -1A00:X04
-LCR:-X1:4 31.6 / X05	X05	BK	0.25 mm²		X05	DBU	0.5 mm²	X05 / 15.4 -1A00:X05
-K4:14 21.7 / X06	X06	DBU	0.5 mm²	4	X06	DBU	0.5 mm²	X06 / 15.4 -1A00:X06
-K5:14 21.9 / X07	X07	DBU	0.5 mm²		X07	DBU	0.5 mm²	X07 / 15.5 -1A00:X07
				5	X10	DBU	0.5 mm²	X10 / 15.6 -1A00:X10
					X11	DBU	0.5 mm²	X11 / 15.6 -1A00:X11
-1X100:1 32.1 / X12	X12	BK/1	1 mm²	6	X12	DBU	0.5 mm²	X12 / 15.7 -1A00:X12
-1X100:2 32.1 / X13	X13	BK/2	1 mm²		X13	DBU	0.5 mm²	X13 / 15.7 -1A00:X13
-1X100:3 32.1 / X14	X14	BK/3	1 mm²	7	X14	DBU	0.5 mm²	X14 / 15.8 -1A00:X14
-1X100:4 32.2 / X15	X15	BK/4	1 mm²		X15	DBU	0.5 mm²	X15 / 15.8 -1A00:X15
-1X100:5 32.2 / X16	X16	BK/5	1 mm²	8	X16	DBU	0.5 mm²	X16 / 15.8 -1A00:X16
-1X100:6 32.2 / X17	X17	BK/6	1 mm²		X17	DBU	0.5 mm²	X17 / 15.9 -1A00:X17
-1X100:7 32.3 / X20	X20	BK/7	1 mm²	9	X20	DBU	0.5 mm²	X20 / 17.1 -1A01:X20
-1X100:8 32.3 / X21	X21	BK/8	1 mm²		X21	DBU	0.5 mm²	X21 / 17.2 -1A01:X21
-1X100:9 32.3 / X22	X22	BK/9	1 mm²	10	X22	DBU	0.5 mm²	X22 / 17.2 -1A01:X22
-1X100:10 32.3 / X23	X23	BK/10	1 mm²		X23	DBU	0.5 mm²	X23 / 17.3 -1A01:X23
-1X100:11 32.4 / X24	X24	BK/11	1 mm²	11	X24	DBU	0.5 mm²	X24 / 17.3 -1A01:X24
-1X100:12 32.4 / X25	X25	BK/12	1 mm²		X25	DBU	0.5 mm²	X25 / 17.3 -1A01:X25
-1X100:13 32.4 / X26	X26	BK/13	1 mm²	12	X26	DBU	0.5 mm²	X26 / 17.4 -1A01:X26
-1X100:14 32.5 / X27	X27	BK/14	1 mm²		X27	DBU	0.5 mm²	X27 / 17.4 -1A01:X27
-1X100:15 32.5 / X30	X30	BK/15	1 mm²	13	X30	DBU	0.5 mm²	X30 / 17.6 -1A01:X30
					X31	DBU	0.5 mm²	X31 / 17.6 -1A01:X31
				14	X32	DBU	0.5 mm²	X32 / 17.6 -1A01:X32
					X33	DBU	0.5 mm²	X33 / 17.7 -1A01:X33
					X34	DBU	0.5 mm²	X34 / 17.7 -1A01:X34
				15	X35	DBU	0.5 mm²	X35 / 17.7 -1A01:X35
					X36	DBU	0.5 mm²	X36 / 17.8 -1A01:X36
				16	X37	DBU	0.5 mm²	X37 / 17.8 -1A01:X37

Modification	REVISION		
Date			
Name			
Original	Ed.	Date	11-04-2024
	Appr.		MANORAJ.SELVARAJ

Replacement of

Replaced by

ROBIS

S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1

Motherson Automotive & Technology engineering

INPUT TERMINAL DISTRIBUTION

FMSPM00857

= / + CP

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OUTPUT TERMINAL DISTRIBUTION

ALL FUSE SHOULD BE 0.5 A

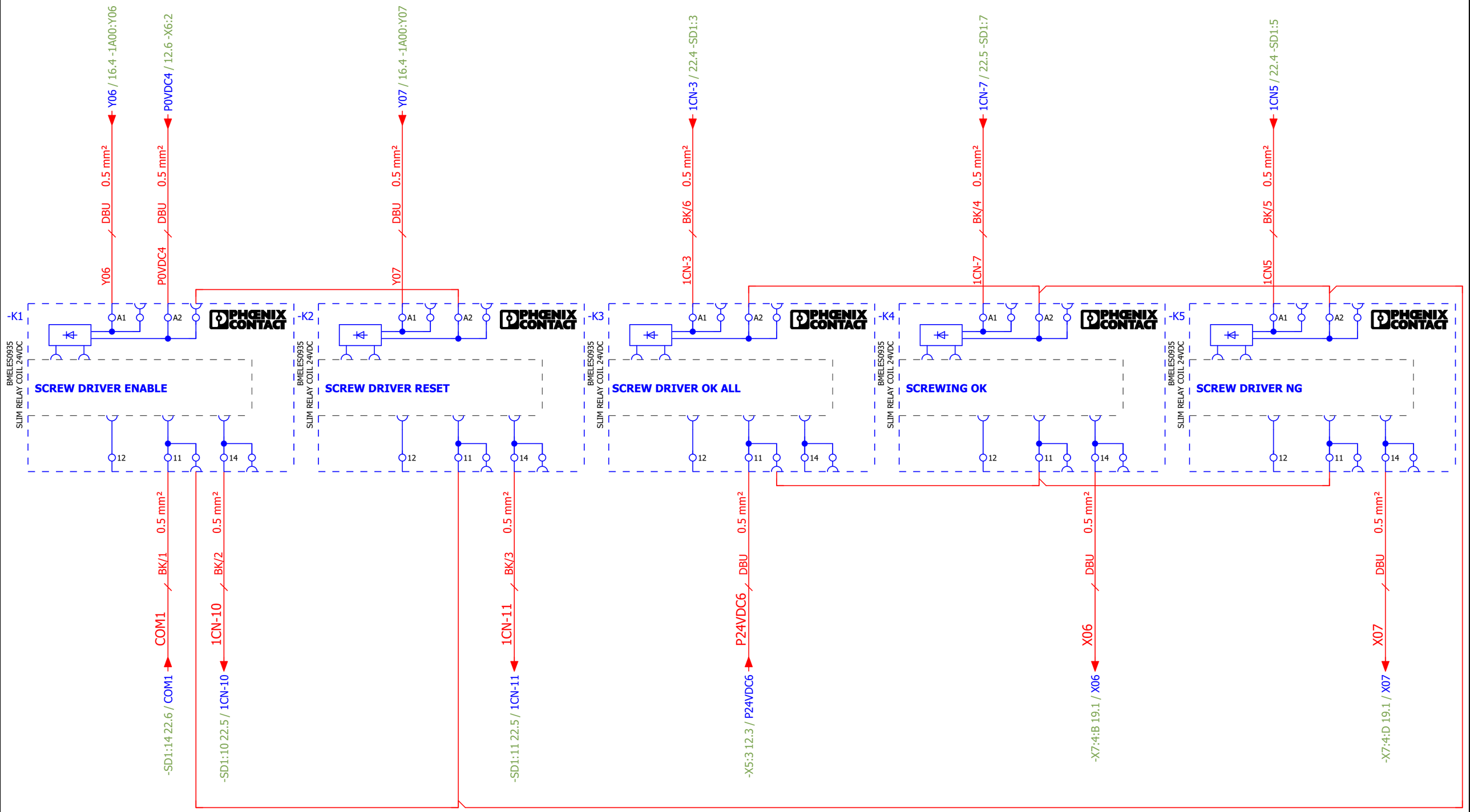


-X8
BMELES2968
TERMINAL BLOCK 2.5 SQ MM
1 IN 2 OUT

Y00	Y01	Y02	Y03	Y04	Y10	Y11	Y12	Y13	Y14	Y15	Y16	Y17	Y20	Y21	Y22	Y23	Y24	Y25	Y26	Y27	Y30	Y31	Y32	Y33	Y34	Y35	Y36	Y37	
DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	
0.5 mm²																													
16.1 -1A00:Y00	16.2 -1A00:Y01	16.2 -1A00:Y02	16.3 -1A00:Y03	16.3 -1A00:Y04	16.6 -1A00:Y10	16.6 -1A00:Y11	16.7 -1A00:Y12	16.7 -1A00:Y13	16.8 -1A00:Y14	16.8 -1A00:Y15	16.8 -1A00:Y16	16.9 -1A00:Y17	18.1 -1A02:Y20	18.2 -1A02:Y21	18.2 -1A02:Y22	18.3 -1A02:Y23	18.3 -1A02:Y24	18.3 -1A02:Y25	18.4 -1A02:Y26	18.4 -1A02:Y27	18.6 -1A02:Y30	18.6 -1A02:Y31	18.6 -1A02:Y32	18.7 -1A02:Y33	18.7 -1A02:Y34	18.8 -1A02:Y35	18.8 -1A02:Y36	18.8 -1A02:Y37	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	
OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	
0.5 mm²																													
BK/1	BK/2	BK/3	BK/4	BK/1	BK/2	BK/3	BK/4	BK/5	BK/6	BK/7	BK/8	BK/9	DBU																
Y00	Y01	Y02	Y03	Y04	Y12	Y13	Y14	Y15	Y16	Y17	Y20	Y21	Y22																
-TL1.0:1 30.8 / Y00	-TL1.0:2 30.8 / Y01	-TL1.0:3 30.8 / Y02	-TL1.0:4 30.9 / Y03	+PP-1X1:1 +PP/34.1 / Y04	+PP-1X1:2 +PP/34.1 / Y12	+PP-1X1:3 +PP/34.2 / Y13	+PP-1X1:4 +PP/34.2 / Y14	+PP-1X1:5 +PP/34.2 / Y15	+PP-1X1:6 +PP/34.3 / Y16	+PP-1X1:7 +PP/34.3 / Y17	+PP-1X1:8 +PP/34.3 / Y20	+PP-1X1:9 +PP/34.4 / Y21	-K7:A1 26.3 / Y22																

REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	OUTPUT TERMINAL DISTRIBUTION		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 20
Modification	Date	Name	Original		Replacement of	Replaced by			Page 20 / 37

RELAY INTERFACE FOR SCREWING DRIVER 1

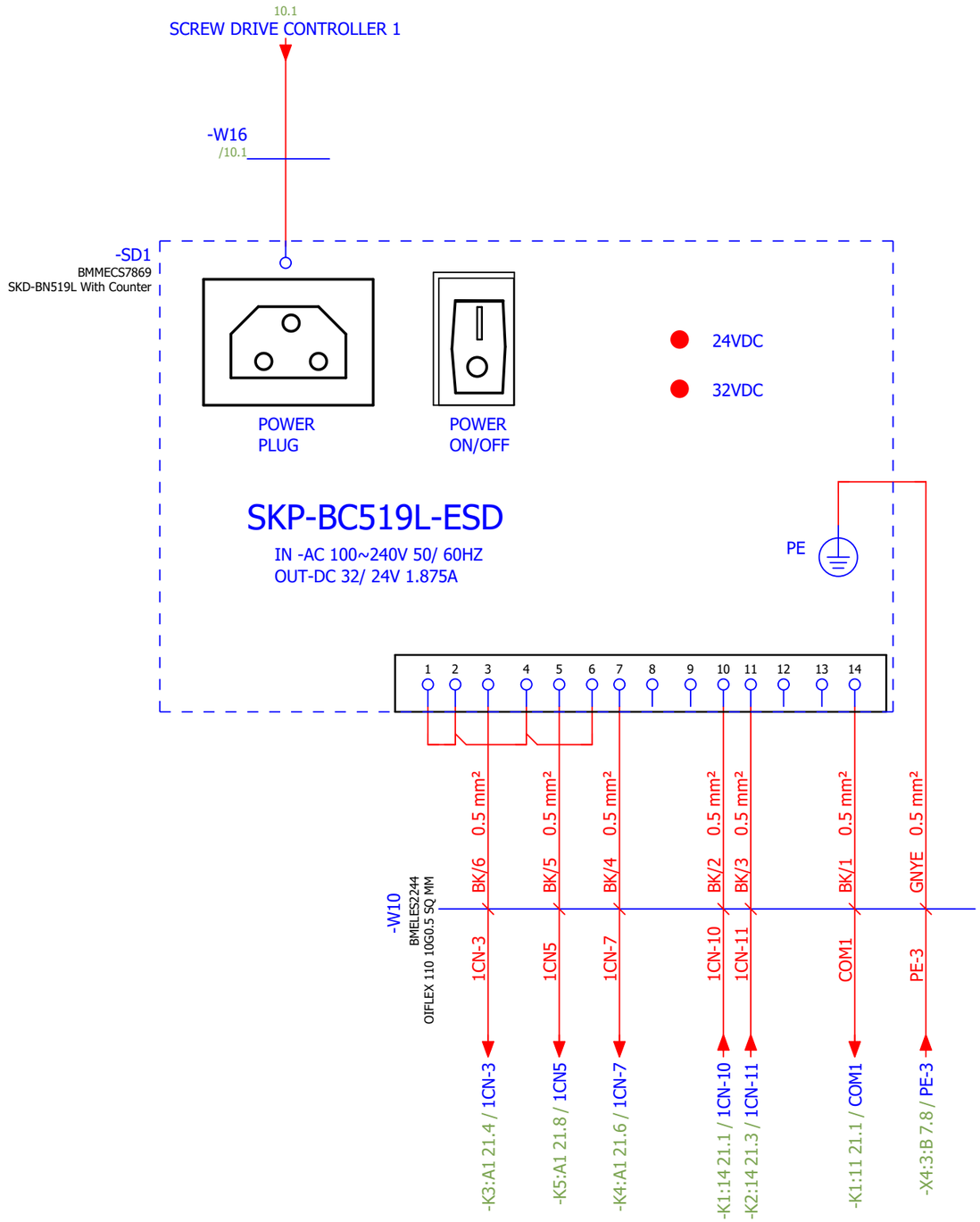


20

22

REVISION			Date	11-04-2024	ROBIS		Motherson Automotive & Technology engineering	RELAY INTERFACE FOR SCREWING DRIVER 1	=
			Ed.	MANORAJ.SELVARAJ					+ CP
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SCREWING DRIVER INTERFACE 1



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			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 22
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IPC INTERFACE



-IPC
BMCONS0728
MOOTEK Industrial PC
MBOX-520-I5-M7-8265U

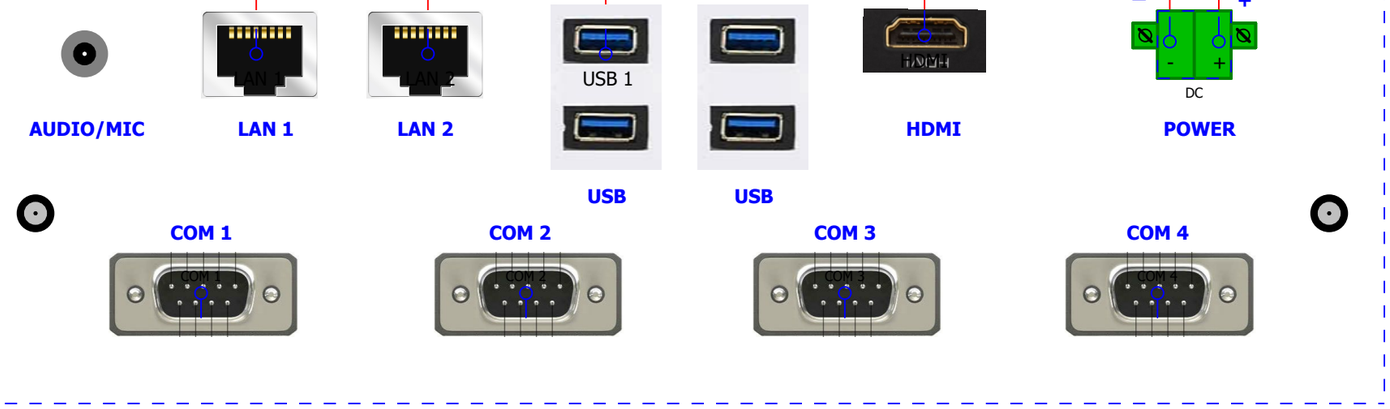
-WC3
BMELES1072
UNITRONIC LAN 600
WC3 / 13.2 -A1:3:3

-WC3.1
BMELES1072
UNITRONIC LAN 600
WC3.1 / 24.1 -MVC-1:1

-WC10
BMELES2202
Male toFemale USB A
Cable , 5mtr(USB 2.0)
TOUCH / 25.4 -MONITOR-TOUCH

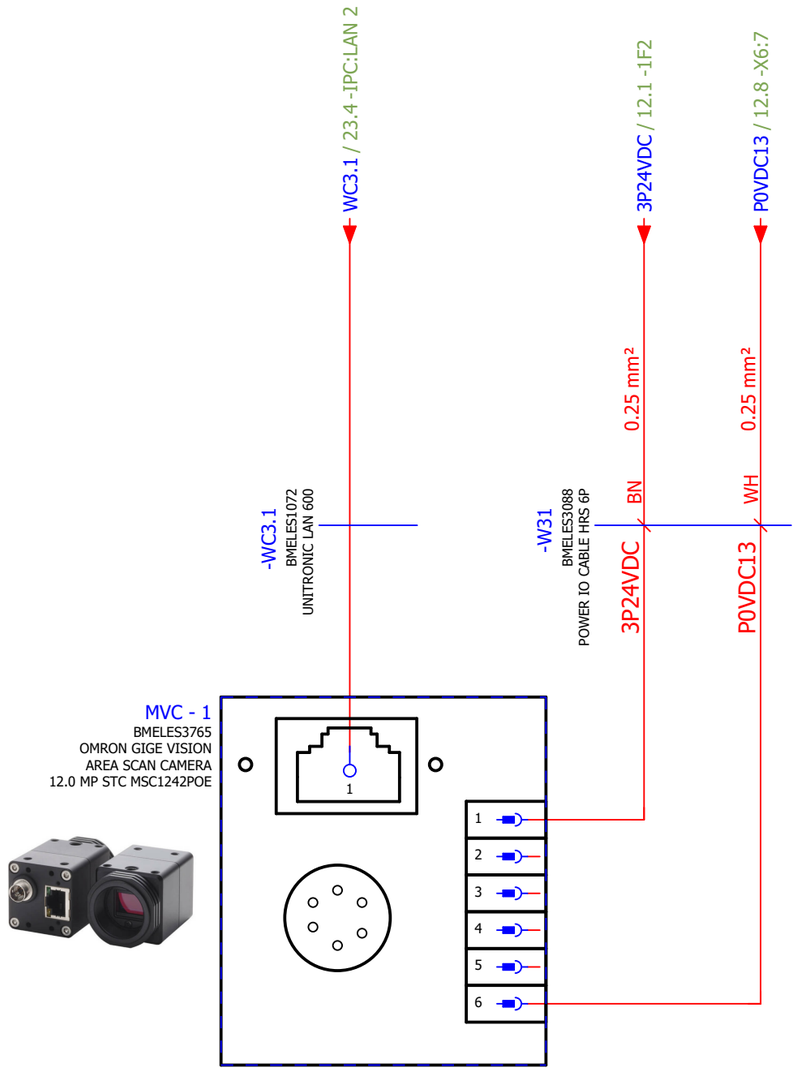
-WC9
HDMI CABLE
HDMI / 25.6 -MONITOR-HDMI

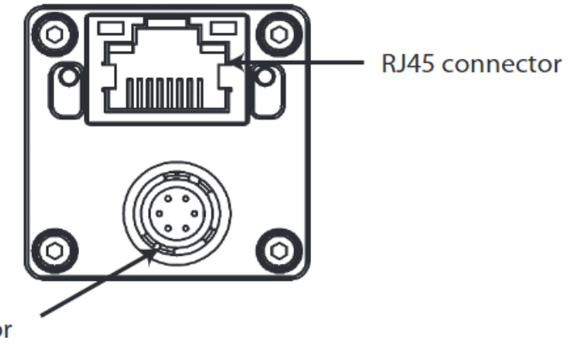
-W30
BMELES1938
3G1.5 sqmm cable
P0VDC12 BK/2 1.5 mm²
2P24VDC BK/1 1.5 mm²
P0VDC12 / 12.8 -LCT:-X1:3
2P24VDC / 12.0 -1F2



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			Ed.	MANORAJ.SELVARAJ					+ CP
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CAMERA INTERFACE





Pin No.	Signal name	I/O	Signal voltage
1	POWER IN	IN	10.8 to 26.4 Vdc
2	Isolated input (Line0)	IN	Low: 1.0 V or less High: 3.0 to 26.4 V * Potential difference between isolated input and isolated I/O common
3	Non-isolated I/O (Line2)	IN / OUT	3.0 to 26.4 V/Open Collector
4	Isolated output (Line1)	OUT	Open Collector
5	Isolated I/O common	IN	
6	GND	IN	0 V

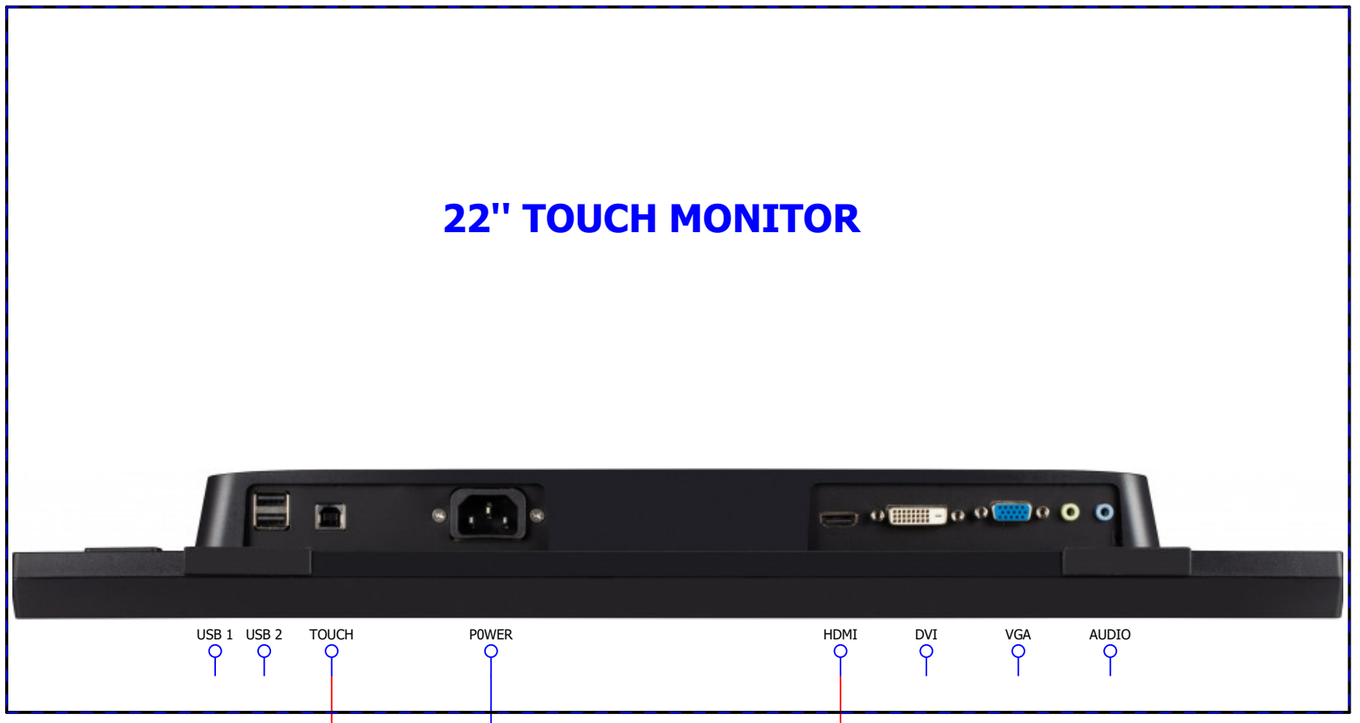


REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	CAMERA INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
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MONITOR INTERFACE

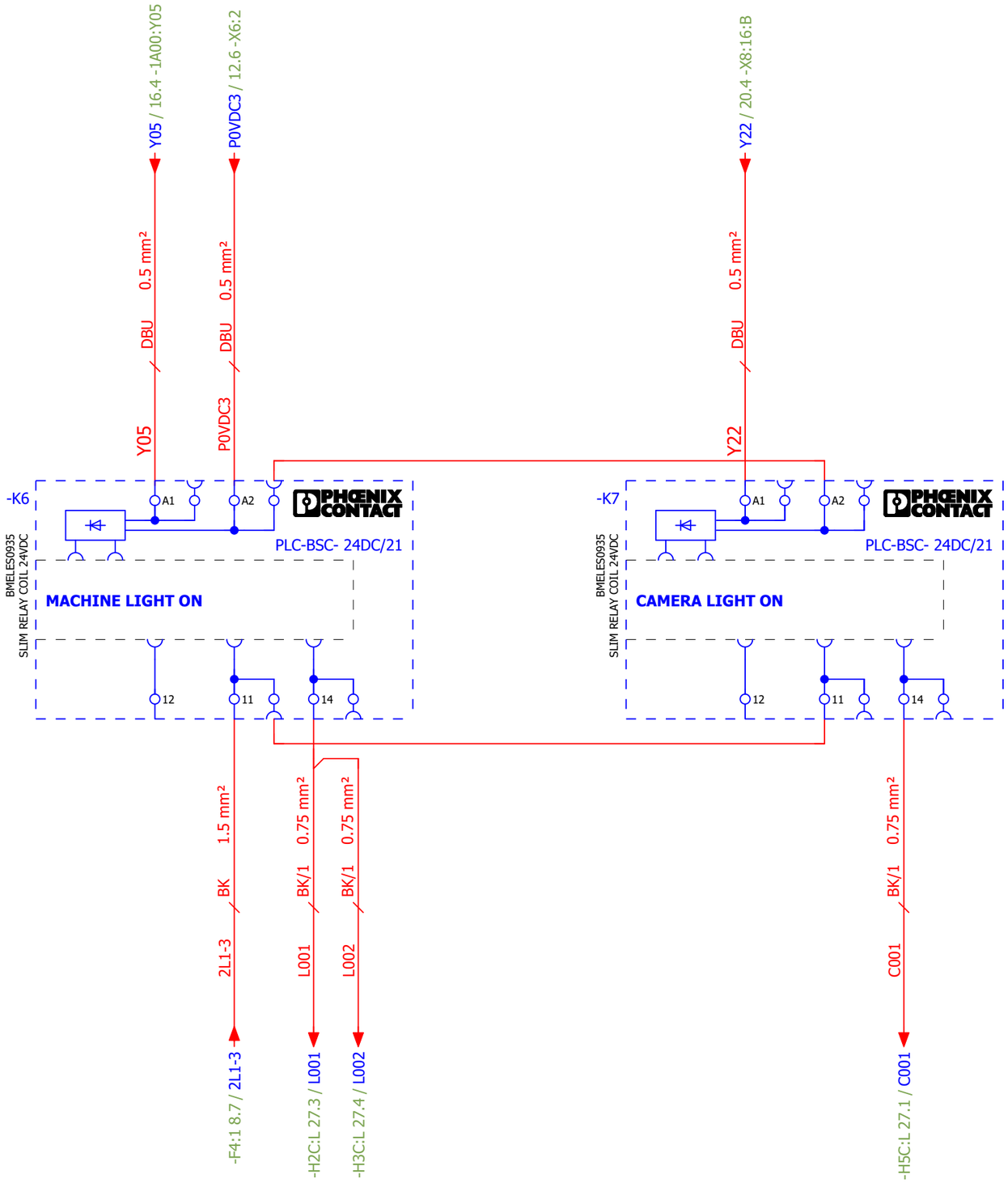
-MONITOR
BMELES3308
22"Touch Monitor





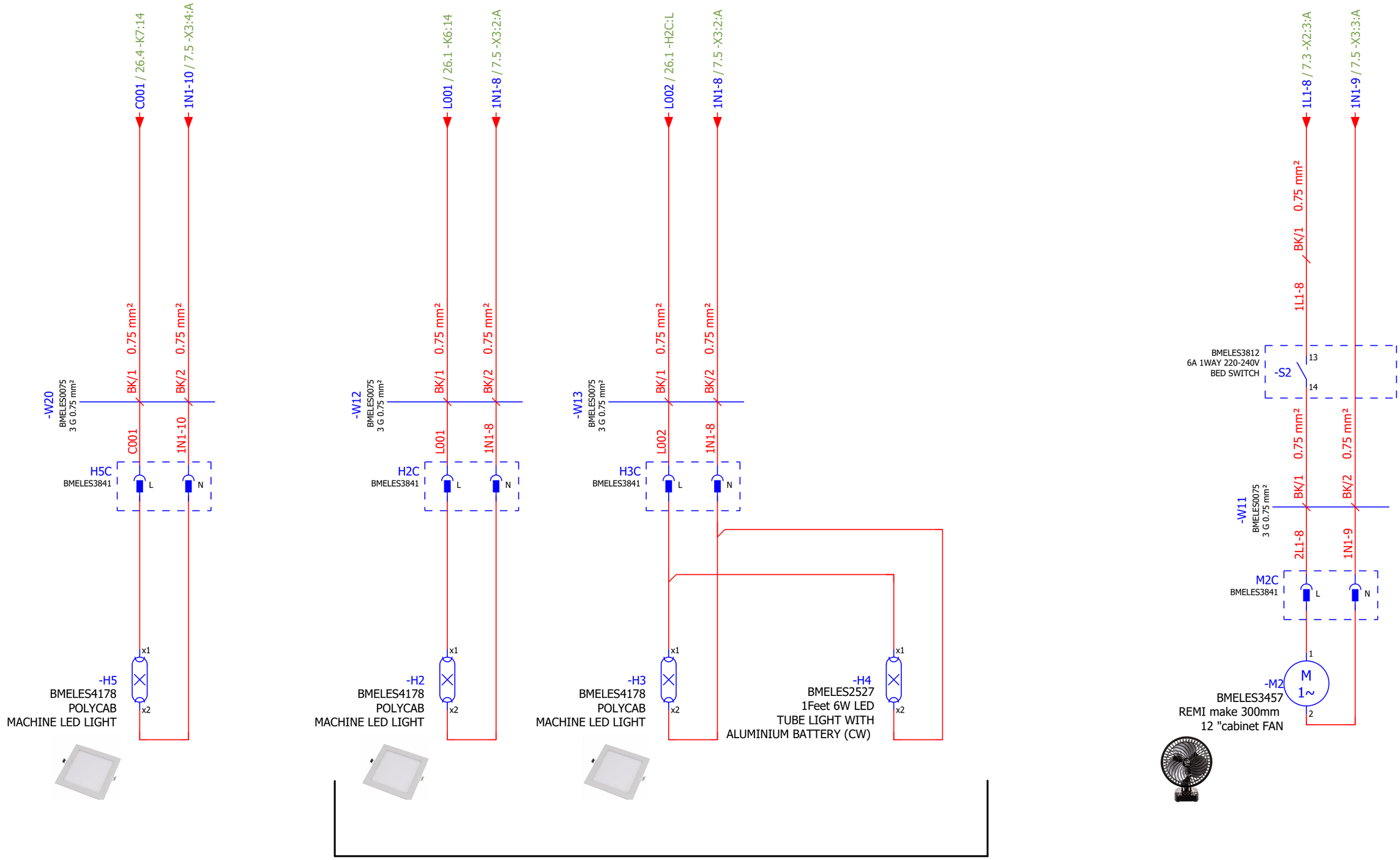
REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	MONITOR INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 25
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MACHINE LIGHT RELAY INTERFACE



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			Ed.	MANORAJ.SELVARAJ					+ CP
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MACHINE LIGHT & FAN INTERFACE



CAMERA LIGHT

MACHINE LIGHT

MACHINE FAN

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			Ed.	MANORAJ.SELVARAJ	S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1				+ CP
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HMI INTERFACE



-WC2 / 13.2 -A1:2:2

-WC6 / 29.2 -HMI RS232-COM1

-WC1
BMELES1072
UNITRONIC LAN 600

-WC6
BMELES0221
0034504 - Unitronic
LIYCY 4 X 0.34

-HMI
CAAAI00000000586
HMI GS2107-WTBD

Ethernet | RS-232 | RS-422 | USB

MITSUBISHI ELECTRIC

\+ | - | FG

0.75 mm² | 0.75 mm² | 0.75 mm²

-W5
BMELES1937
3 G 0.75 mm²

BK/1 | BK/2 | GNYE

1P24VDC | P0VDC6 | PE-8

-1F1 / 12.0 / 1P24VDC

-X6:3 / 12.7 / P0VDC6

-X4:2:A / 7.7 / PE-8

HMI POWER SUPPLY

P24VDC12 / 12.4 -X5:6

P0VDC5 / 12.7 -X6:3

X04 / 19.1 -X7:3:B

-W6
BMELES1937
3 G 0.75 mm²

0.75 mm² | 0.75 mm² | 0.75 mm²

BK/1 | BK/3 | BK/2

P24VDC12 | P0VDC5 | X04

-PSC
3 PIN CONNECTOR

BN | BU | BK | WH

PS1

1 2 3 4

P

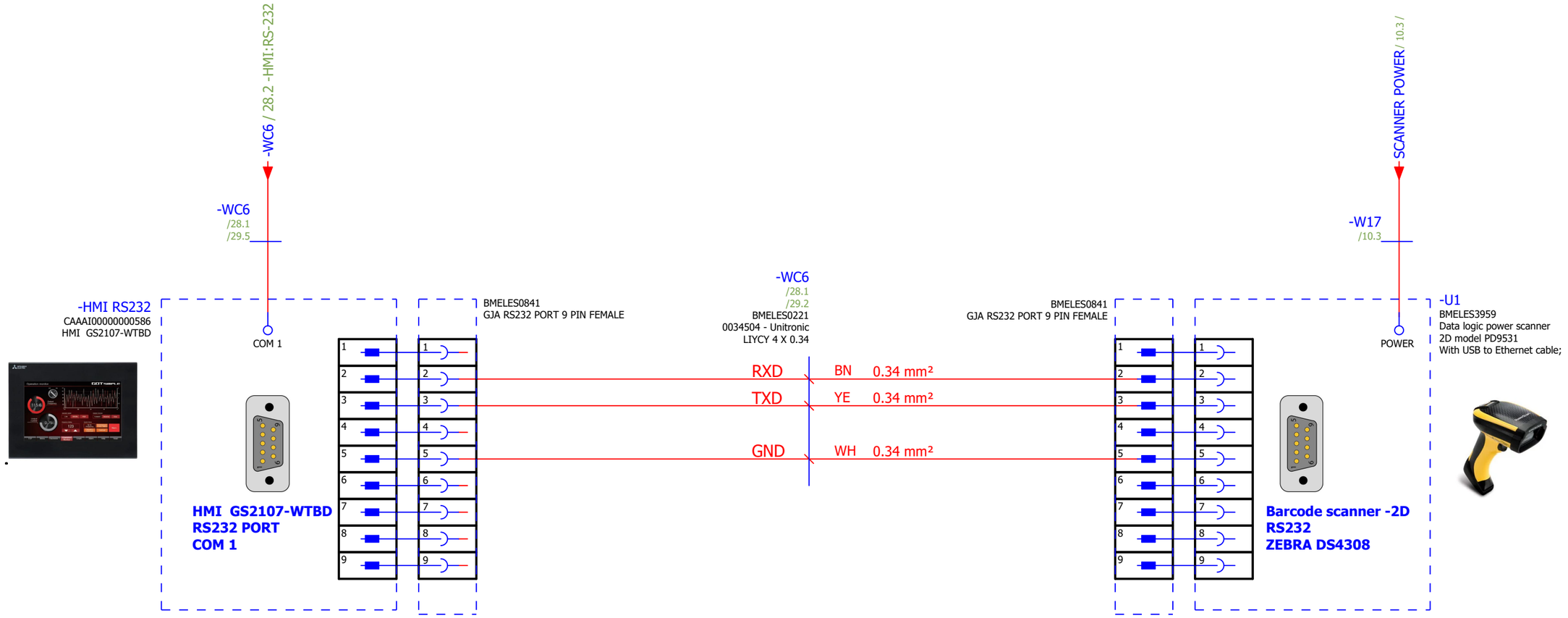


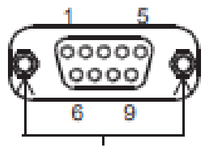
PRESSURE SWITCH

REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	HMI INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 28
Modification	Date	Name	Original		Replacement of	Replaced by			Page 28 / 37

0	1	2	3	4	5	6	7	8	9

HMI TO SCANNER COMMUNICATION



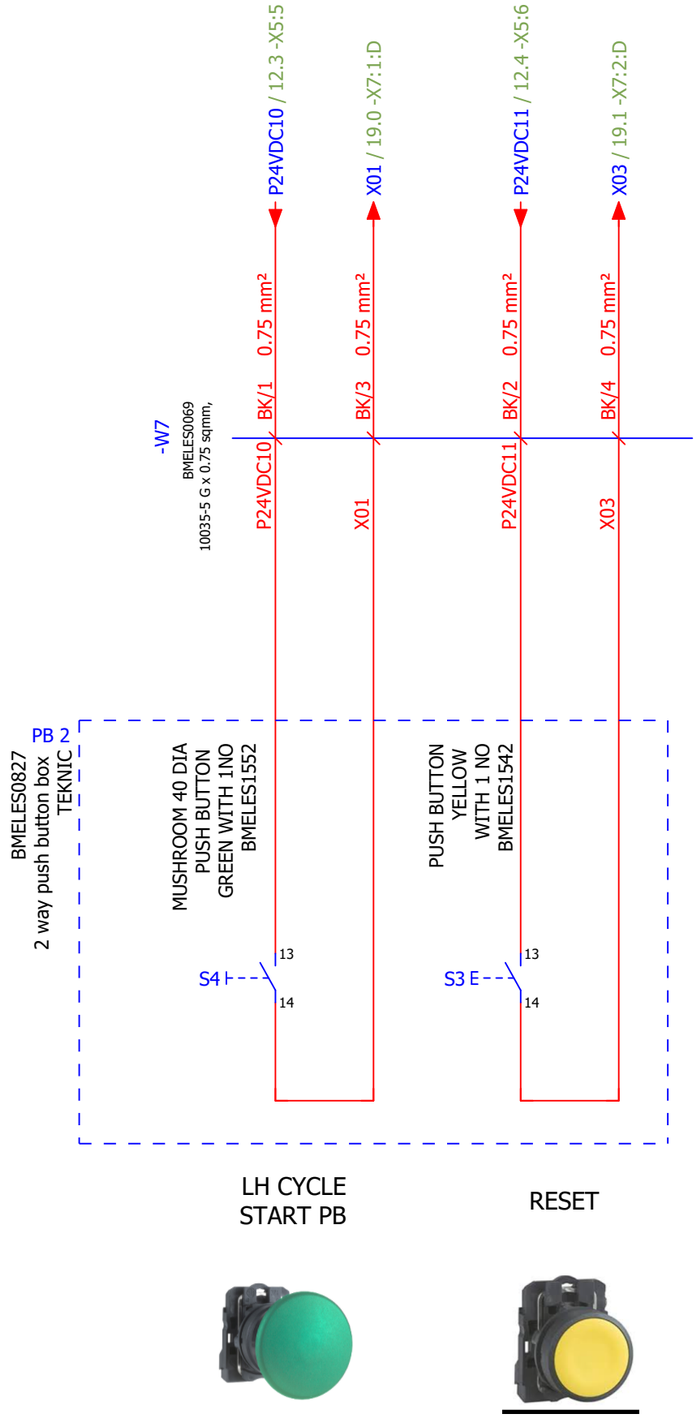


Screws to fix
RS-232C connector
#4-40UNC
(inch screw thread)

Pin No.	Signal	Name
1	CD (DCD)	Receive carrier detection
2	RD (RXD)	Receive data
3	SD (TXD)	Send data
4	ER (DTR)	Send request
5	SG (GND)	Signal ground
6	DR (DSR)	Send enabled
7, 8, 9		Not used

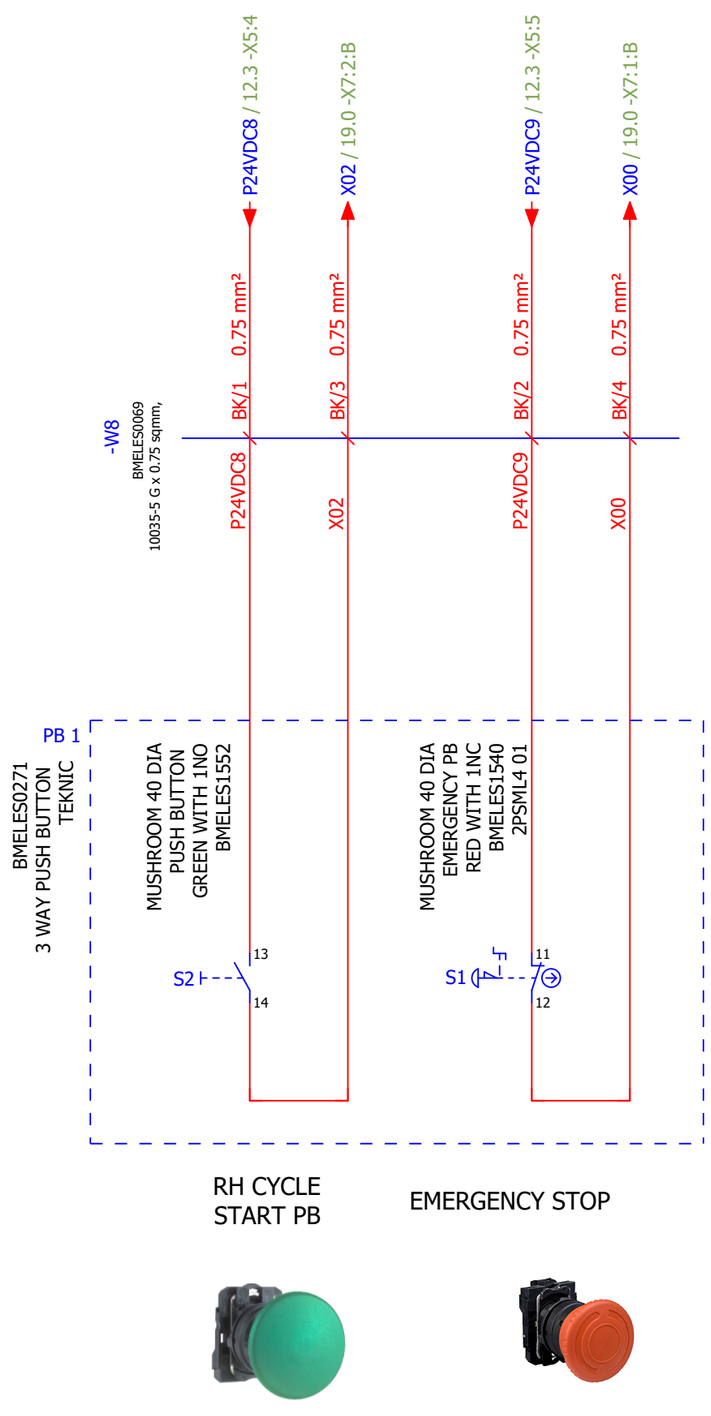
REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	HMI TO SCANNER COMMUNICATION		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 29
Modification	Date	Name	Original		Replacement of	Replaced by			Page 29 / 37

PUSHBUTTON & TOWER LAMP INTERFACE



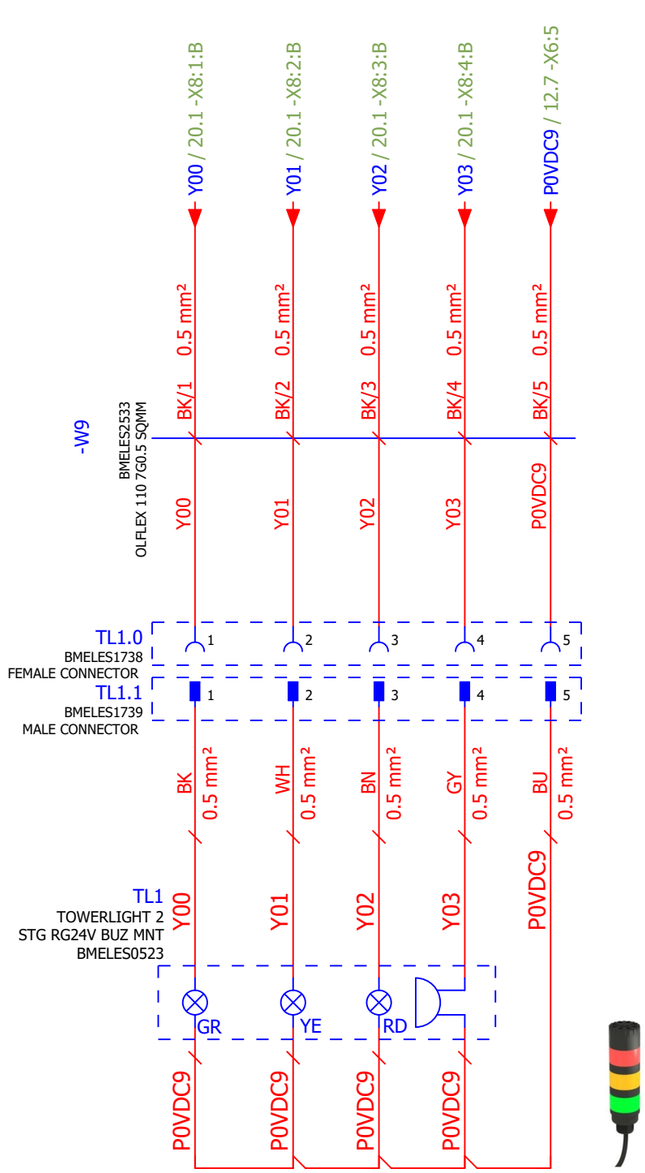
LH CYCLE START PB

RESET



RH CYCLE START PB

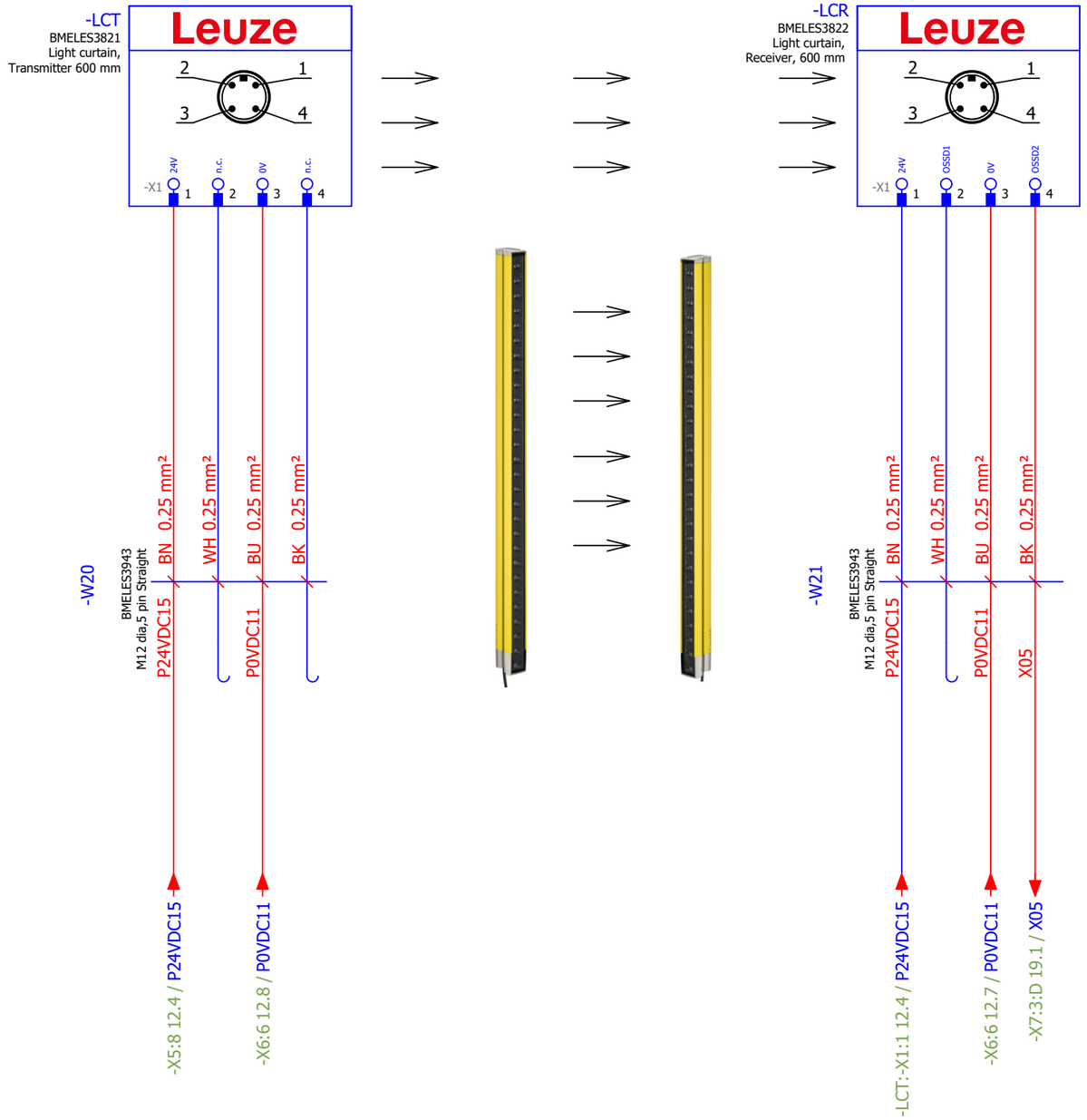
EMERGENCY STOP



TOWER LAMP

REVISION			Date	11-04-2024	ROBIS		Motherson Automotive & Technology engineering	PUSHBUTTON & TOWER LAMP INTERFACE	=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1				FMSPM00857 Page 30
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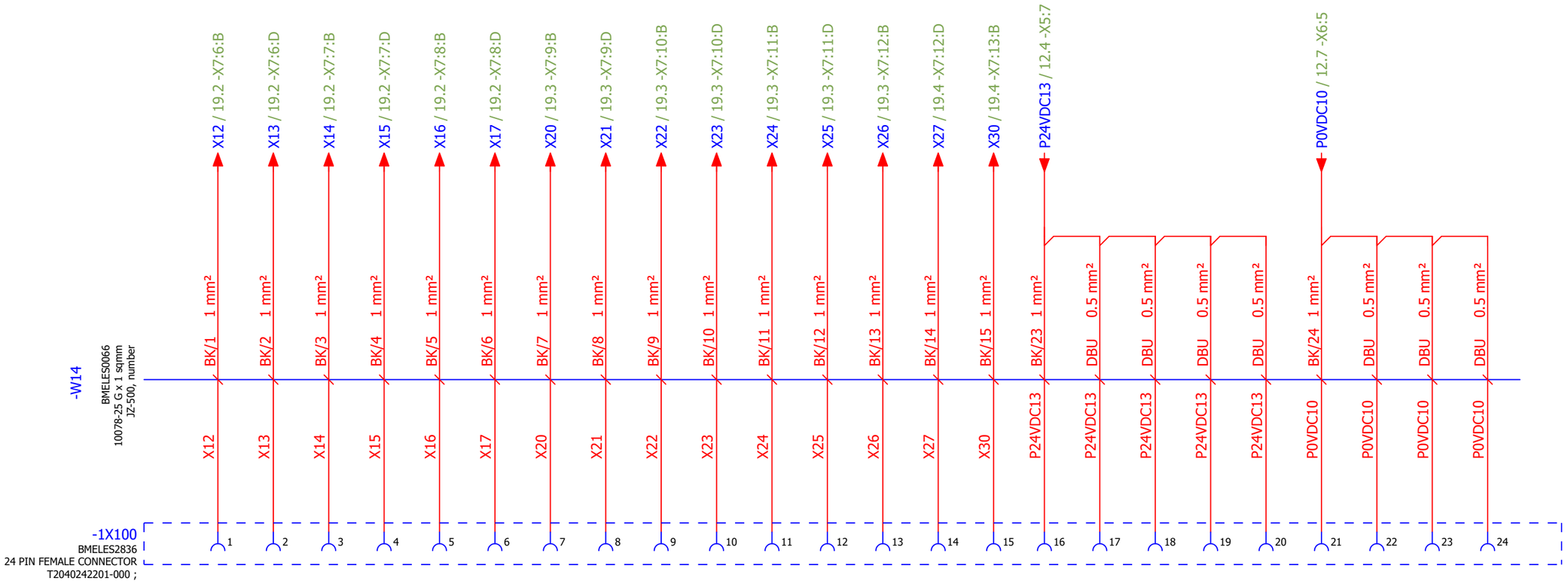
LIGHT CURTAIN & DOOR LIMIT SWITCH INTERFACE



SAFETY LIGHT CURTAIN

REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	LIGHT CURTAIN & DOOR LIMIT SWITCH INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 31
Modification	Date	Name	Original		Replacement of	Replaced by			Page 31 / 37

24 PIN I/O INTERFACE

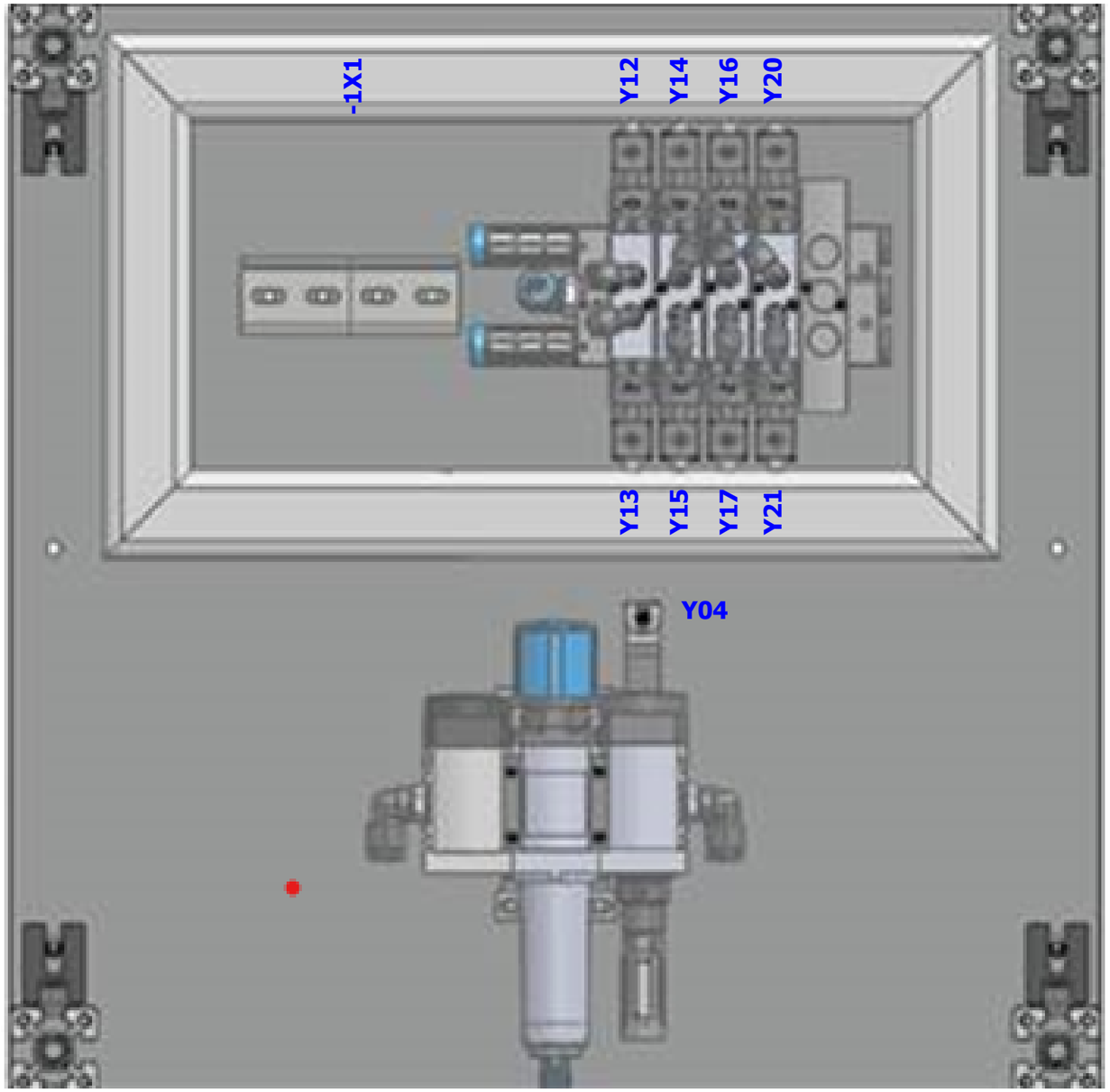


31

+PP/33

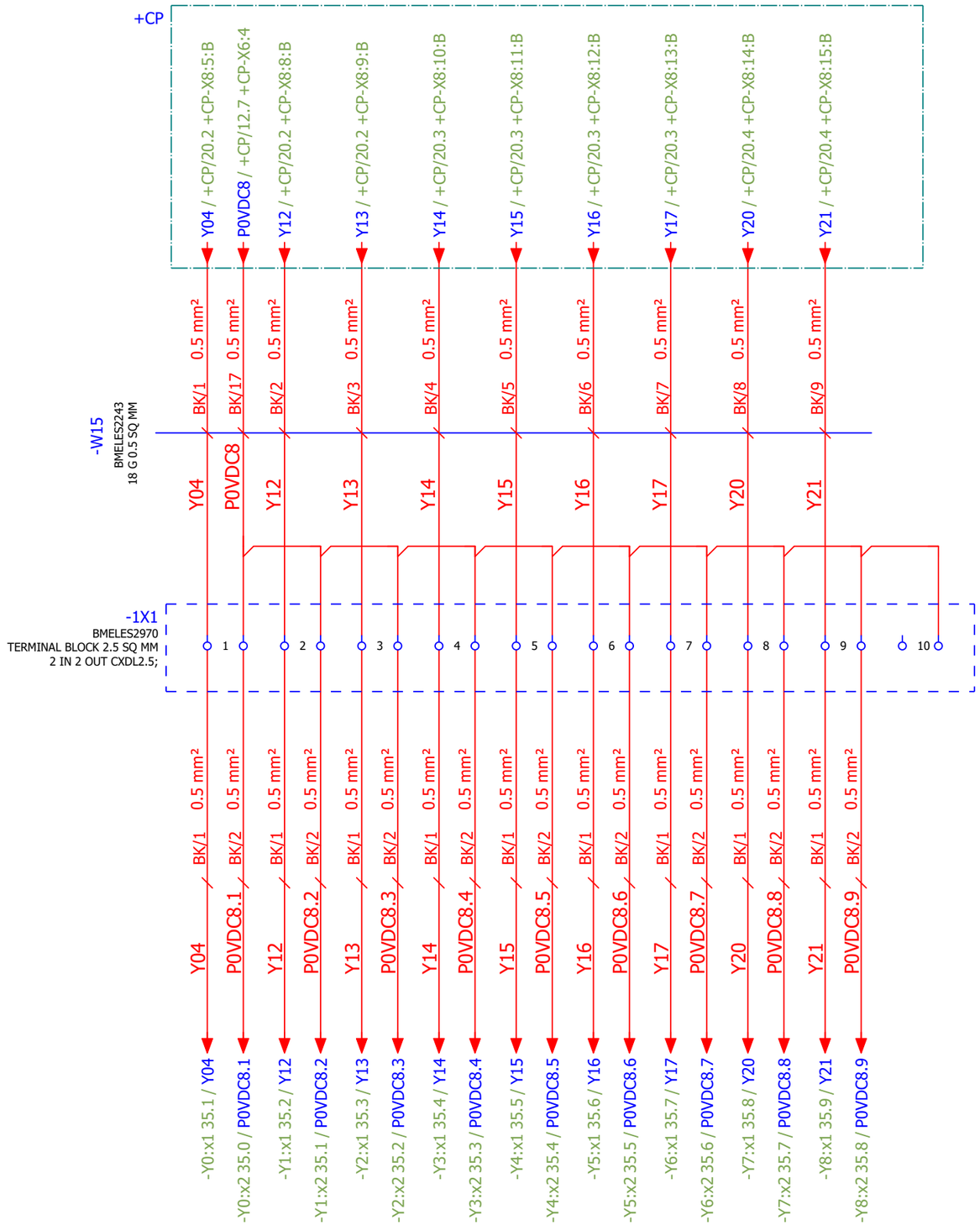
REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	24 PIN I/O INTERFACE	=	
			Ed.	MANORAJ.SELVARAJ				+ CP	
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 32
Modification	Date	Name	Original		Replacement of	Replaced by			Page 32 / 37

CUSTOMIZE PANEL



TAG NO	DESCRIPTION
Y04	SHUT OFF VALVE
Y12	CLAMPING CYL1 FWD SOL
Y13	CLAMPING CYL1 REV SOL
Y14	SPARE VALVE
Y15	SPARE VALVE
Y16	SPARE VALVE
Y17	SPARE VALVE
Y20	SPARE VALVE
Y21	SPARE VALVE
-1X1	BMELES2970 TERMINAL BLOCK 2.5 SQ MM 2 IN 2 OUT CXDL2.5;

PNEUMATIC TERMINAL BLOCK



REVISION			Date	11-04-2024	ROBIS		Motherson Automotive & Technology engineering	PNEUMATIC TERMINAL BLOCK	=
			Ed.	MANORAJ.SELVARAJ					+ PP
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 34
Modification	Date	Name	Original		Replacement of	Replaced by			Page 34 / 37

SHUT OFF VALVE

Y04
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.1 BK/2 0.5 mm² → P0VDC8.1 / 34.1 -1X1
-Y0 x2 x1
Y04 BK/1 0.5 mm² → Y04 / 34.1 -1X1:1

CLAMP CYL 1 FWD SOL

Y12
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.2 BK/2 0.5 mm² → P0VDC8.2 / 34.1 -1X1
-Y1 x2 x1
Y12 BK/1 0.5 mm² → Y12 / 34.1 -1X1:2

CLAMP CYL 1 REV SOL

Y13
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.3 BK/2 0.5 mm² → P0VDC8.3 / 34.2 -1X1
-Y2 x2 x1
Y13 BK/1 0.5 mm² → Y13 / 34.2 -1X1:3

SPARE VALVE

Y14
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.4 BK/2 0.5 mm² → P0VDC8.4 / 34.2 -1X1
-Y3 x2 x1
Y14 BK/1 0.5 mm² → Y14 / 34.2 -1X1:4

SPARE VALVE

Y15
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.5 BK/2 0.5 mm² → P0VDC8.5 / 34.3 -1X1
-Y4 x2 x1
Y15 BK/1 0.5 mm² → Y15 / 34.2 -1X1:5

SPARE VALVE

Y16
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.6 BK/2 0.5 mm² → P0VDC8.6 / 34.3 -1X1
-Y5 x2 x1
Y16 BK/1 0.5 mm² → Y16 / 34.3 -1X1:6

SPARE VALVE

Y17
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.7 BK/2 0.5 mm² → P0VDC8.7 / 34.3 -1X1
-Y6 x2 x1
Y17 BK/1 0.5 mm² → Y17 / 34.3 -1X1:7

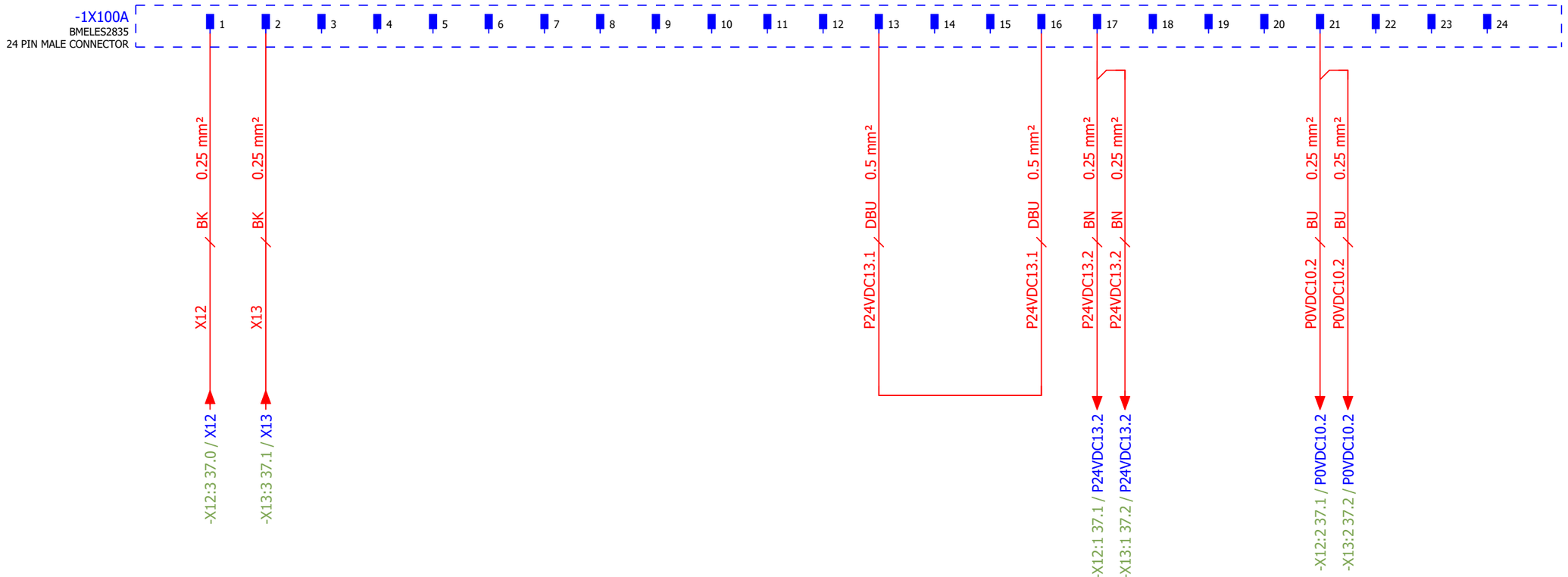
SPARE VALVE

Y20
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.8 BK/2 0.5 mm² → P0VDC8.8 / 34.4 -1X1
-Y7 x2 x1
Y20 BK/1 0.5 mm² → Y20 / 34.3 -1X1:8

SPARE VALVE

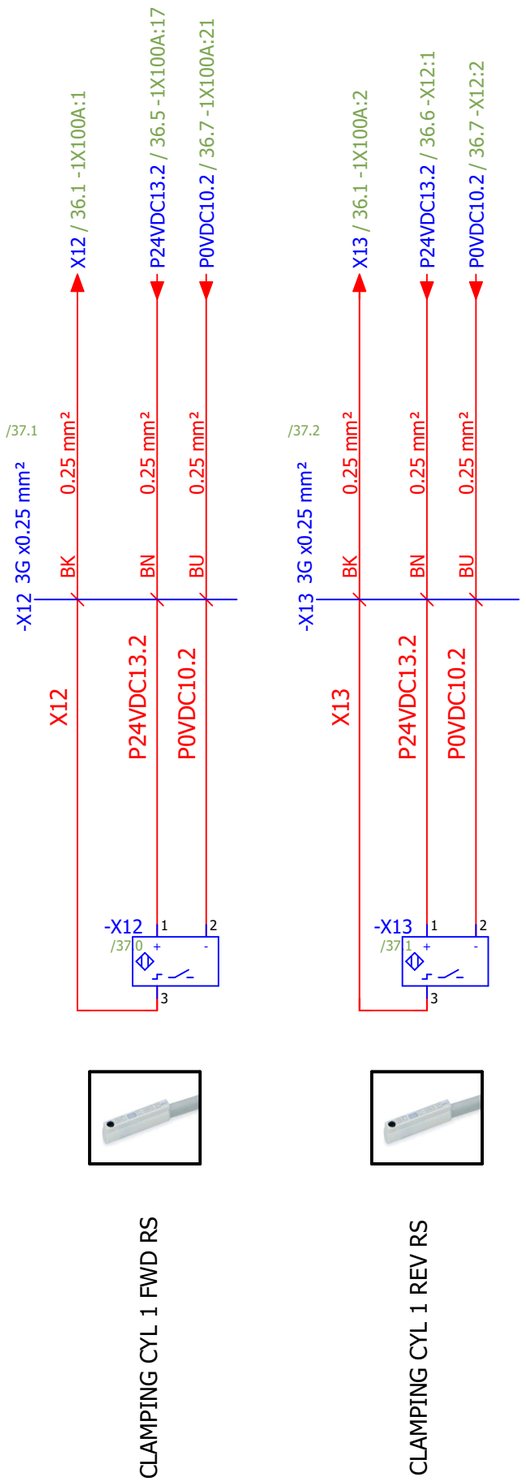
Y21
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.9 BK/2 0.5 mm² → P0VDC8.9 / 34.4 -1X1
-Y8 x2 x1
Y21 BK/1 0.5 mm² → Y21 / 34.4 -1X1:9

24 PIN I/O INTERFACE



REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	24 PIN I/O INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ FIX
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 36
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REED SWITCH INTERFACE



REVISION			Date	11-04-2024	ROBIS		Motherson Automotive & Technology engineering	REED SWITCH INTERFACE	=
			Ed.	MANORAJ.SELVARAJ					+ FIX
			Appr.		S220 HL HV & LV OUTERLENS TP BEZEL SCREWING OFFLINE-1			FMSPM00857	Page 37
Modification	Date	Name	Original		Replacement of	Replaced by			Page 37 / 37